Adversarial Machine Learning

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08 - 2026

Robust optimization

General robust optimization problem

$$\min_{w \in \Omega} \max_{u \in U} f(w, u)$$

- Decision w: What we control: weights, power, schedule, beamformer, model parameters.
- Uncertainty u: What we do not control: channel gain, demand, noise, measurement error, input perturbation.
- Loss $f(w, u)$: Performance under uncertainty u. In ML this is usually a loss function.
- Optimize for the worst allowed case

First-order method

$$\min_{w \in \Omega} \varphi(w)$$

$$\varphi(w) = \max_{u \in U} f(w, u)$$

- At w^k, find a worst case: $u^k \in \arg\max_{u \in U} f(w^k, u)$.
- Update x by projected (sub)gradient descent:

$$w^{k+1} = \mathcal{P}_\Omega \left[w^k - \eta_k \nabla_w f(w^k, u^k) \right]$$

Adversarial Machine Learning

https://adversarial-ml-tutorial.org/

Adversarial examples

- **Clean input:** A normal sample x with label y is correctly classified
- **Small perturbation:** Add δ with small norm, for example $\left|\left|\delta\right|\right|_{\infty} \leq \varepsilon$.
- **Adversarial input:** $x_{adv} = x + \delta$ causes high loss or wrong classification.

Robust training

- Standard training problem:

$$\min_{\theta} \frac{1}{m} \sum_{i=1}^{m} f_{\theta}\,(x_i, y_i)$$

where x_i, y_i are an image and its label; f_θ is the loss function, θ is model parameter.

E.g., $f_\theta(.) = L\big(h_\theta(.)\big)$, where L is the cross-entropy loss and h is the extract feature in the multiclass classification problem.

- Adversarial robust training:

$$\min_{\theta} \max_{\delta \in \Delta} \frac{1}{m} \sum_{i=1}^{m} f_\theta(x_i + \delta, y_i)$$

- **Inner problem:** (attack, adversarial example)

$$\delta^* = \arg\max_{\delta \in \Delta} f_\theta(x_i + \delta, y_i)$$

  - Keep θ fixed. find δ that maximizes the loss for each x_i
  - FGSM and PGD

- **Outer problem:** (adversarial training)
  - Use the adversarial examples found by the inner problem
  - Update θ using SGD/Adam

$$\theta \leftarrow \theta - \eta \nabla_\theta f_\theta(x_i + \delta^*, y_i)$$

Common perturbation sets

ℓ_∞ BALL

Each pixel/feature can change by at most ε. Common for image attacks.

$$\Delta = \{\, \delta : \left|\left|\delta\right|\right|_\infty \leq \varepsilon \,\}$$

ℓ_2 BALL

The total Euclidean perturbation energy is bounded.

$$\Delta = \{\delta : ||\delta||_2 \leq \varepsilon \,\}$$

ℓ_1 BALL

Encourages sparse perturbations, changing fewer coordinates.

$$\Delta = \{\delta : ||\delta||_1 \leq \varepsilon\}$$

Fast Gradient Sign Method (FGSM)

- Goal: for each observation (x, y), $\max_{||\delta||_\infty \leq \epsilon} f_\theta(x+\delta, y)$
- First-order Taylor approximation

$$f_\theta(x+\delta, y) \approx f_\theta(x, y) + \nabla_x f_\theta(x, y)^T \delta$$

- The inner problem becomes a linear maximization over the ε-box

$$\max_{||\delta||_\infty \leq \varepsilon} g^T \delta, \quad \text{where} \quad g = \nabla_x f_\theta(x, \mathrm{y})$$

- $||\delta||_\infty \leq \varepsilon$ means each coordinate satisfies $-\varepsilon \leq \delta_i \leq \varepsilon$. ($i = 1, \ldots, d$. Recall: δ, x have the same dimension d.)

$$\delta_i = \varepsilon \text{ if } g_i \geq 0, \ \delta_i = -\varepsilon \text{ if } g_i < 0$$

$$\rightarrow \delta_i = \varepsilon \operatorname{sign}(g_i).$$

$$\delta_{FGSM} = \varepsilon \operatorname{sign}(\nabla_x f_\theta(x, \mathrm{y}))$$

FGSM algorithm

- Input: clean example x, label y, model f_θ, perturbation budget ε.
- Forward pass: compute loss $f_\theta(x, y)$.
- Backward pass: compute input gradient

$$g = \nabla_x f_\theta(x, y)$$

- Create perturbation: $\delta = \varepsilon \, \mathrm{sign}(g)$.
- Create adversarial example: $x_{adv} = clip(x + \delta)$.

Targeted FGSM

- Untargeted attack: increase the true-label loss

$$\delta = +\varepsilon\, sign\left(\nabla_x f_\theta(x, y_{true})\right)$$

- Targeted attack: make the model predict target class y_{target}

$$\delta = -\varepsilon\, sign(\nabla_x f_\theta(x, y_{target})$$

Why FGSM can be weak?

- FGSM uses first-oder Taylor's approximation of the loss around x .
- Deep networks are nonlinear; the gradient can change inside the ε-ball.
- A one-step attack may under-estimate the true worst-case loss.
- For evaluation, a weak attack can give a false sense of robustness

Exercise: derive FGSM for logistic regression

- Binary classifier: $p = \sigma(w^{\mathrm{T}}x + b)$.
- Cross-entropy loss: $L = -y \log p - (1-y)\log(1-p)$.
- Task: compute $\nabla_x L$ and write δ_{FGSM} for $||\delta||_\infty \leq \varepsilon$.

- Expected result

$$\nabla_x L = (p - y)w$$

$$\delta_{FGSM} = \epsilon \,\mathrm{sign}((p - y)w)$$

Projection Gradient Descent (PGD)

- Take many smaller steps and recompute the gradient after each step
- PGD update for a threat model Δ:

$$\delta^{(k+1)} = \mathcal{P}_{\delta \in \Delta}\left[\, \delta^{(k)} + \alpha \nabla_{\delta} f_{\theta}\left(x + \delta^{(k)}, y\right)\right]$$

where $\mathcal{P}$ is the projection onto the ball of interest.

- For an ℓ_∞ ball, projection is coordinate-wise clipping:

$$\mathcal{P}_{\left\{||\delta||_\infty \leq \varepsilon\right\}}(v) = \; clip(v, -\varepsilon, +\varepsilon)$$

PGD for the ℓ_2 threat model

Using Projected normalized steepest ascent method:

$$\delta^{(k+1)} = \mathcal{P}_{||\delta||_2 \leq \varepsilon}\left[\delta^{(k)} + \alpha \frac{\nabla_\delta f_\theta\left(x + \delta^{(k)}, y\right)}{\left|\left|\nabla_\delta f_\theta(x + \delta^{(k)}, y)\right|\right|_2}\right]$$

Projection rescales δ if it leaves the ball

$$\mathcal{P}_{||\delta||_2 \leq \varepsilon}(v) = v \cdot \frac{\varepsilon}{\max\left(\varepsilon, \left||v|\right|_2\right)}$$

PGD adversarial training

- For each image, use PGD to generate adversarial example.
- Then update θ using SGD/Adam on the adversarial loss.

$$\theta \leftarrow \theta - \eta \nabla_\theta f_\theta(x + \delta_{PGD}, y)$$

- This approximates the outer minimization of the robust objective.
- It is stronger but more expensive than FGSM adversarial training

Excercises

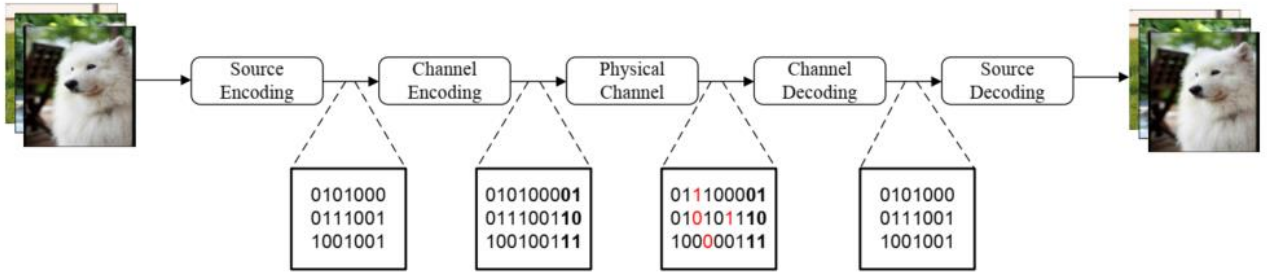
- Run the example code at https://adversarial-ml-tutorial.org/

Robust Semantic Communication

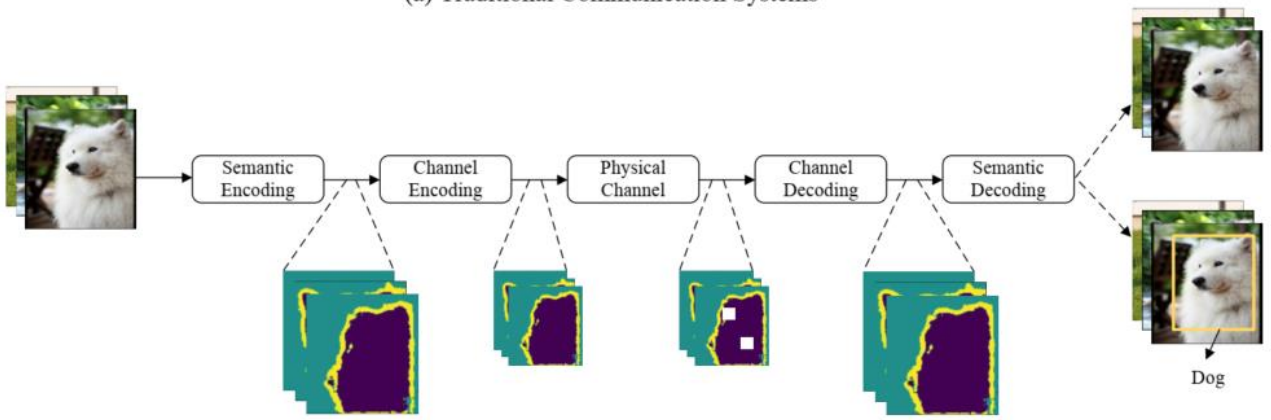
Semantic communication (SemCom)

Traditional communication: transmit bit-by-bit

SemCom: transmit necessary information relevant to the specific task at the receiver



(a) Traditional Communication Systems



(b) Semantic Communication Systems

Semantic communication (SemCom)

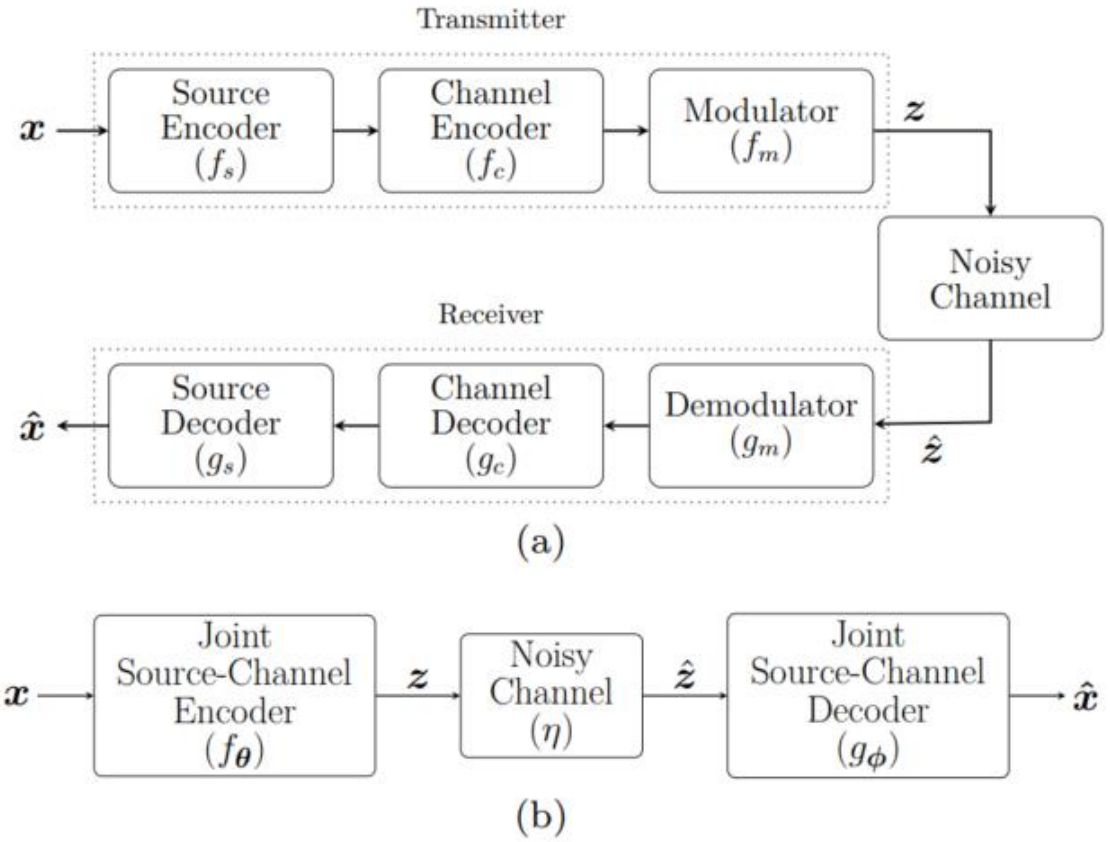
- will be used in internet of things, smart manufacturing, smart transportation, video conference, online education, *etc.*
- Deep learning (DL) and its applications, such as natural language processing (NLP), speech recognition, and computer vision provide significant insights on developing semantic communications

DeepJSCC

- Joint source and channel coding (JSCC) technique for wireless image transmission
- Directly maps the image pixel values to the complex-valued channel input symbols
- Code: https://github.com/chunbaobao/Deep-JSCC-PyTorch

Bourtsoulatze, Eirina, David Burth Kurka, and Deniz Gündüz. "Deep joint source-channel coding for wireless image transmission." *IEEE Transactions on Cognitive Communications and Networking* 5.3 (2019): 567-579.

DeepJSCC

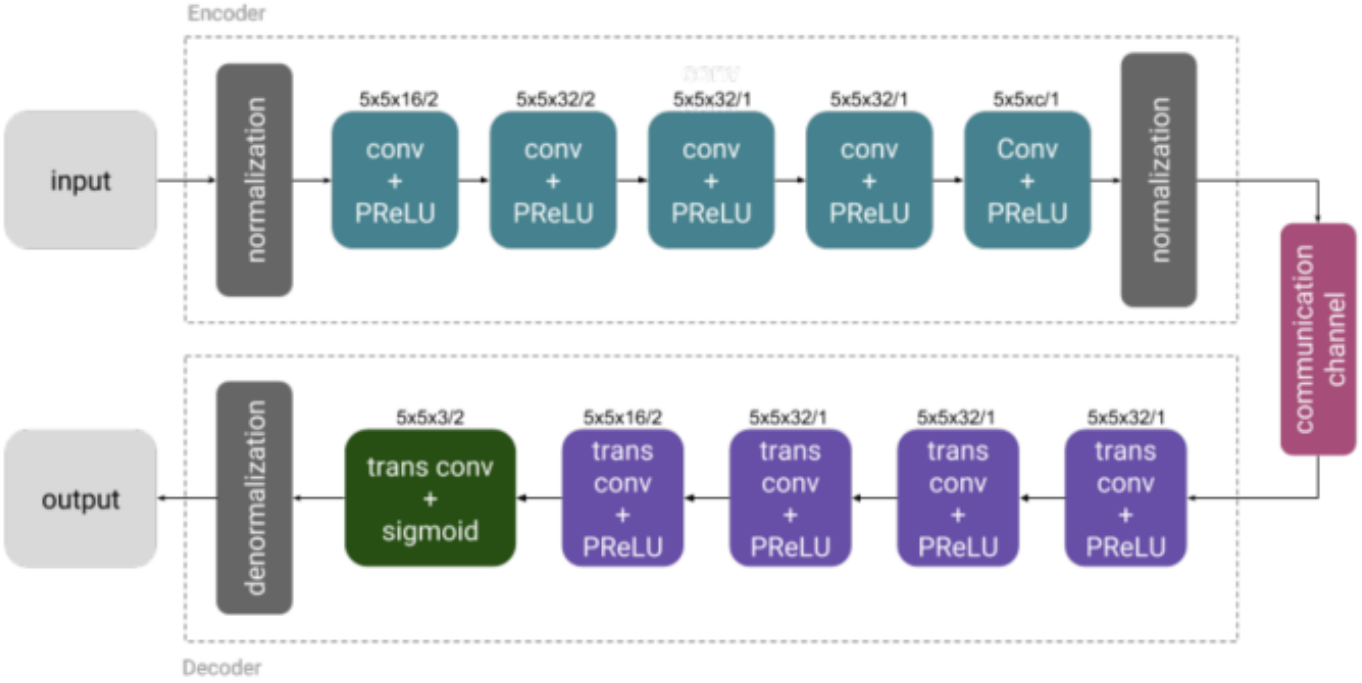


Block diagram of the point-to-point image transmission system:

(a) components of the conventional processing pipeline and

(b) components of the proposed **DeepJSCC** framework.

Architectures of DeepJSCC



- Encodes: a series of convolutional layers followed by parametric ReLU (PReLU) activation functions (ResNet)
- Channel: non-trainable layer, $\hat{z} = \eta(z) = hz + n$
- Decoder: symmetric, transpose CNN

DeepJSCC – Optimization problem

- Optimization problem:

$$(\boldsymbol{\theta}^*, \boldsymbol{\phi}^*) = \arg\min_{\boldsymbol{\theta},\boldsymbol{\phi}} \mathbb{E}_{p(\boldsymbol{x},\hat{\boldsymbol{x}})}[d(\boldsymbol{x}, \hat{\boldsymbol{x}})],$$

- Using empirical mean instead

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N} d(\boldsymbol{x}_i, \hat{\boldsymbol{x}}_i),$$

where $d(x, \hat{x}) = MSE = \frac{1}{n}\left|\left|x - \hat{x}\right|\right|^2$, mean square error of the distortion.

Performance metrics, dataset

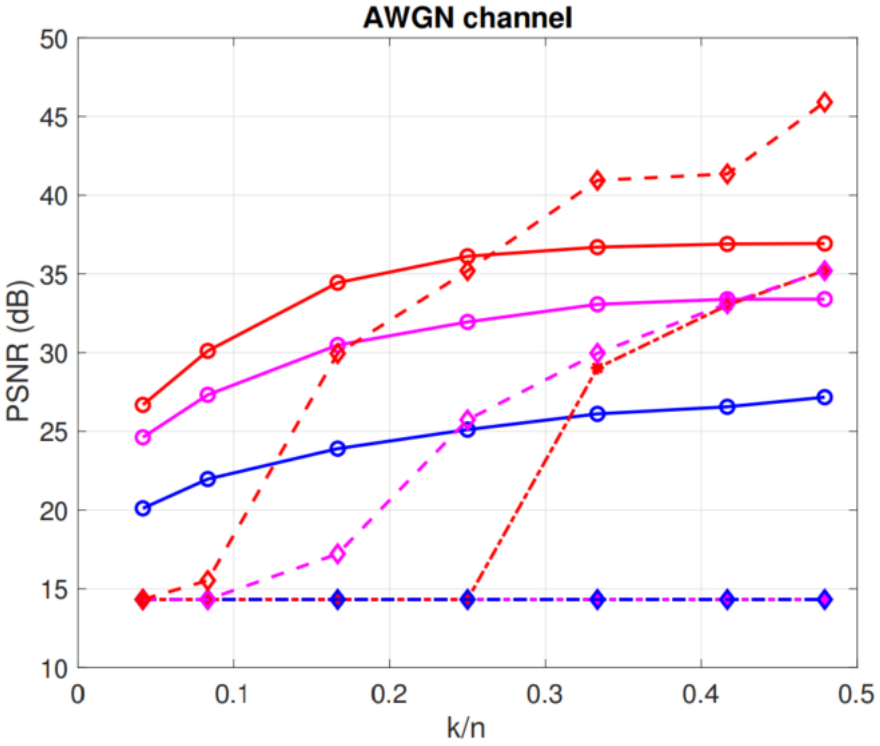
- Peak signal-to-noise ratio (PSNR) metric measures the ratio between the maximum possible power of the signal and the power of the noise that corrupts the signal

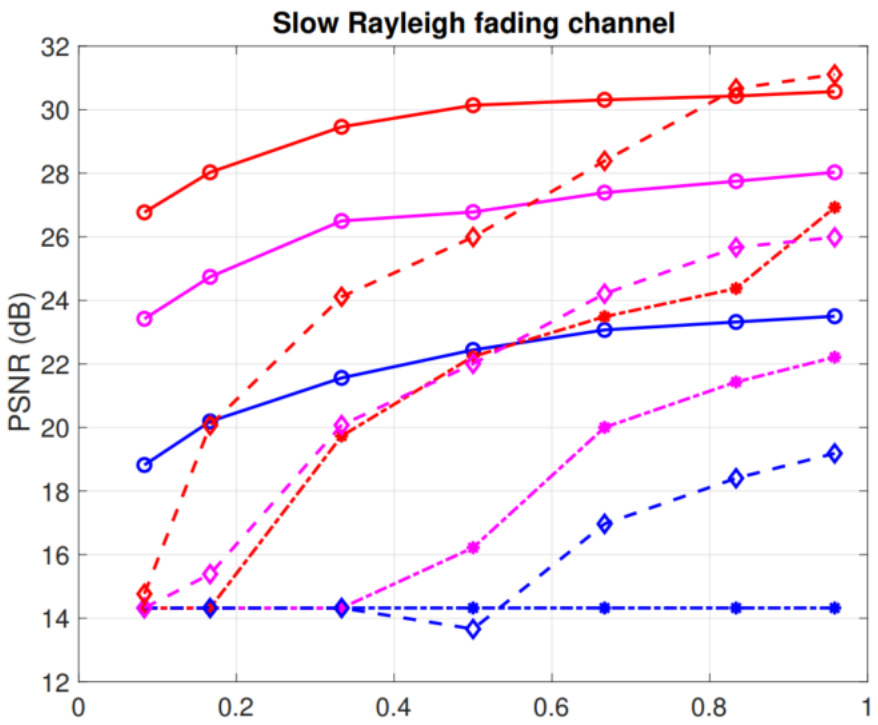
$$\mathrm{PSNR} = 10 \log_{10} \frac{\mathrm{MAX}^2}{\mathrm{MSE}} \quad (\mathrm{dB}).$$

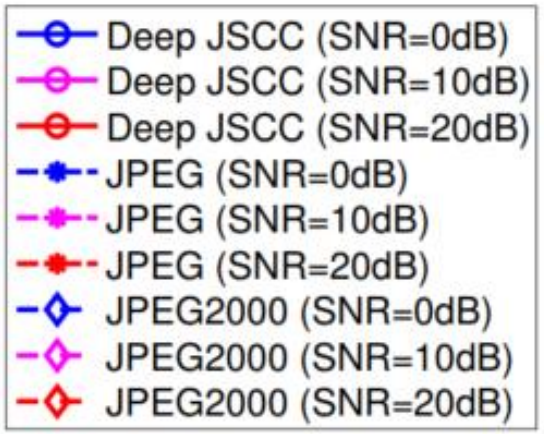
  where MAX is the maximum possible value of the image pixels (255)

- Structural similarity index (SSIM): comparison on luminance, contrast and structure

$$\mathrm{SSIM}(\mathbf{x}, \mathbf{y}) = [l(\mathbf{x}, \mathbf{y})]^{\alpha} \cdot [c(\mathbf{x}, \mathbf{y})]^{\beta} \cdot [s(\mathbf{x}, \mathbf{y})]^{\gamma}$$

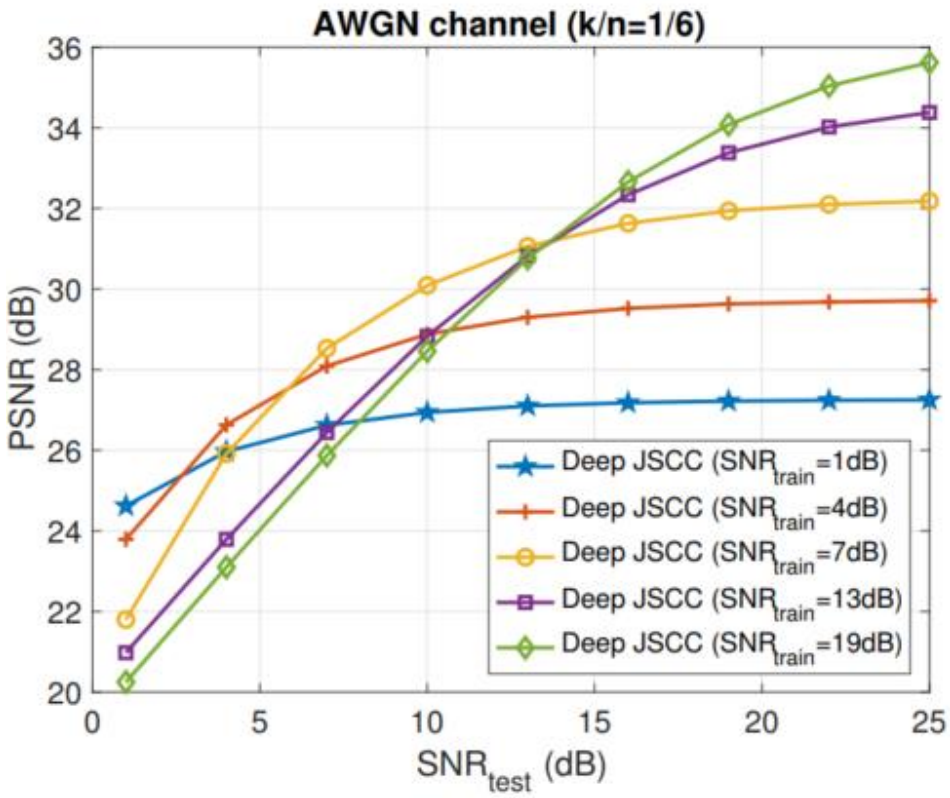


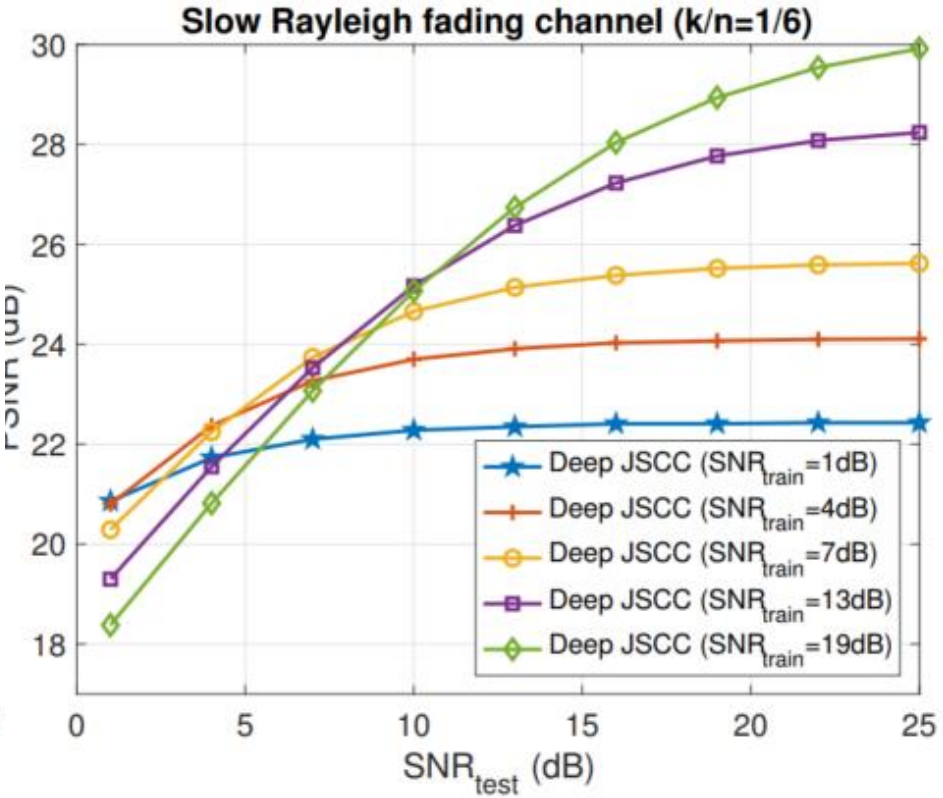




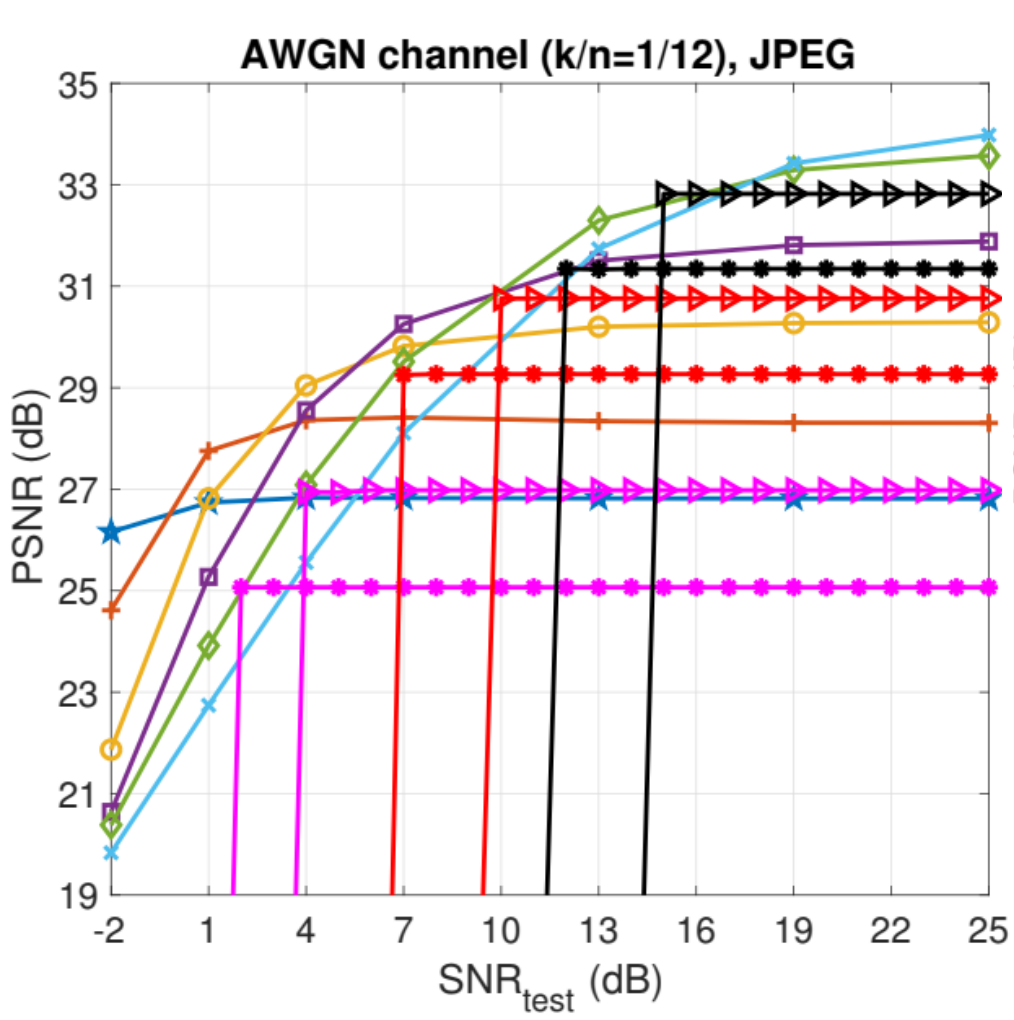
DeepJSCC on CIFAR-10

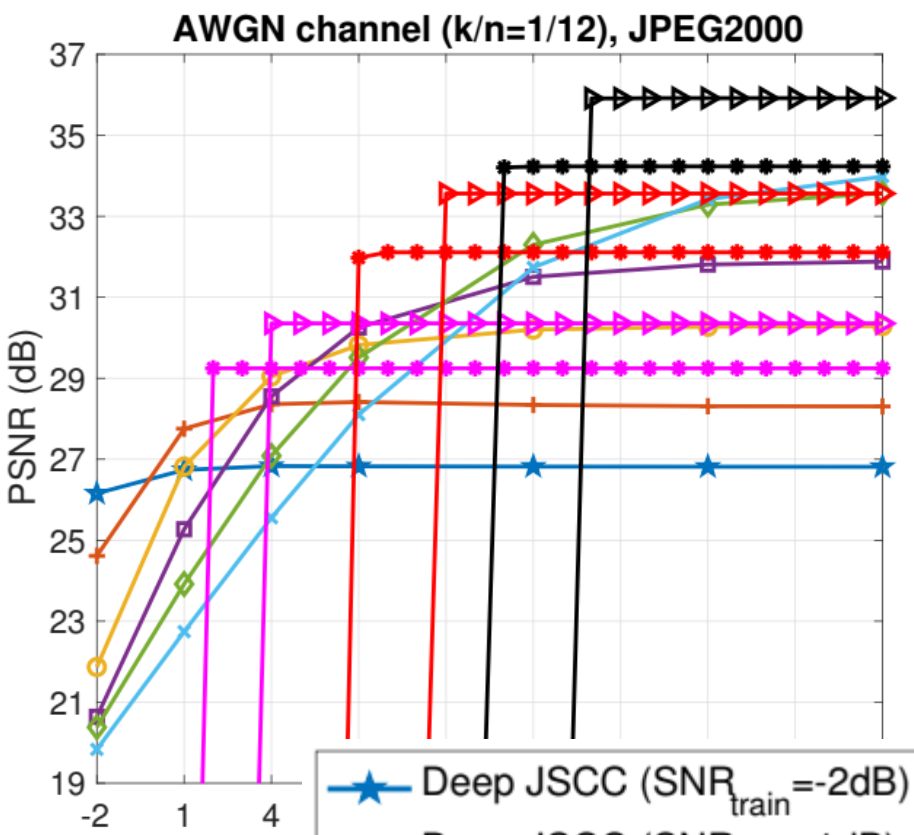
- PSNR vs. bandwidth compression ratio, k/n, for different SNR values.
- For each case, the same target SNR value is used in training and evaluation.
- DeepJSCC is the best at low k/n

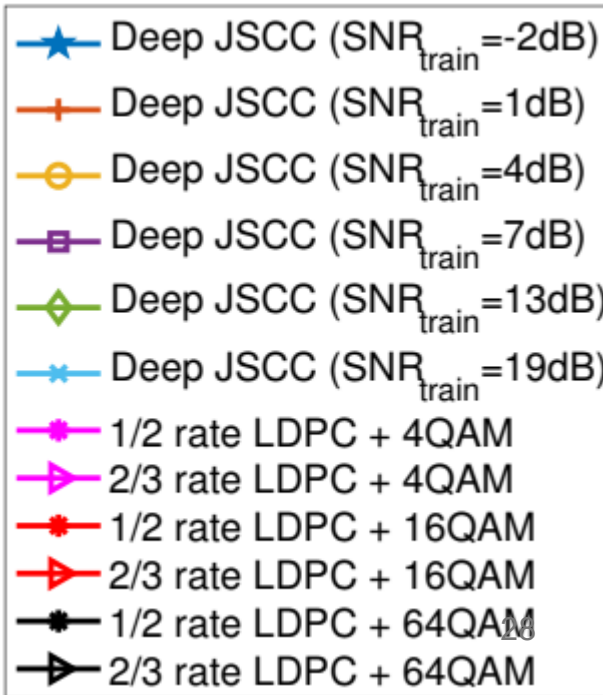




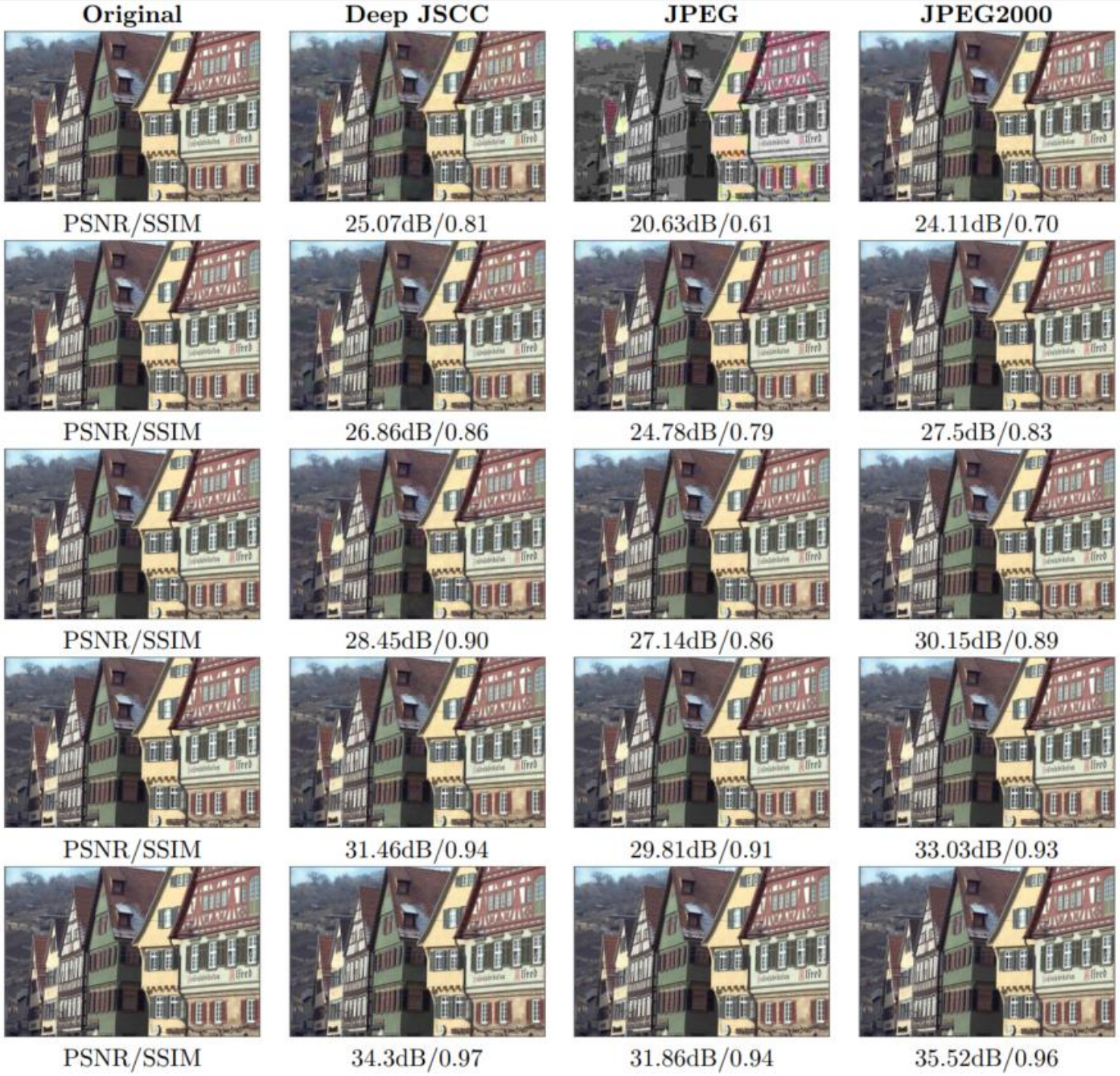
- CIFAR-10 dataset
- PSNR vs SNR







- Kodak dataset
- JPEG, JPEG2000 source coding schemes
- DeepJSCC: best at low SNR, no “cliff-effect”



SNRs

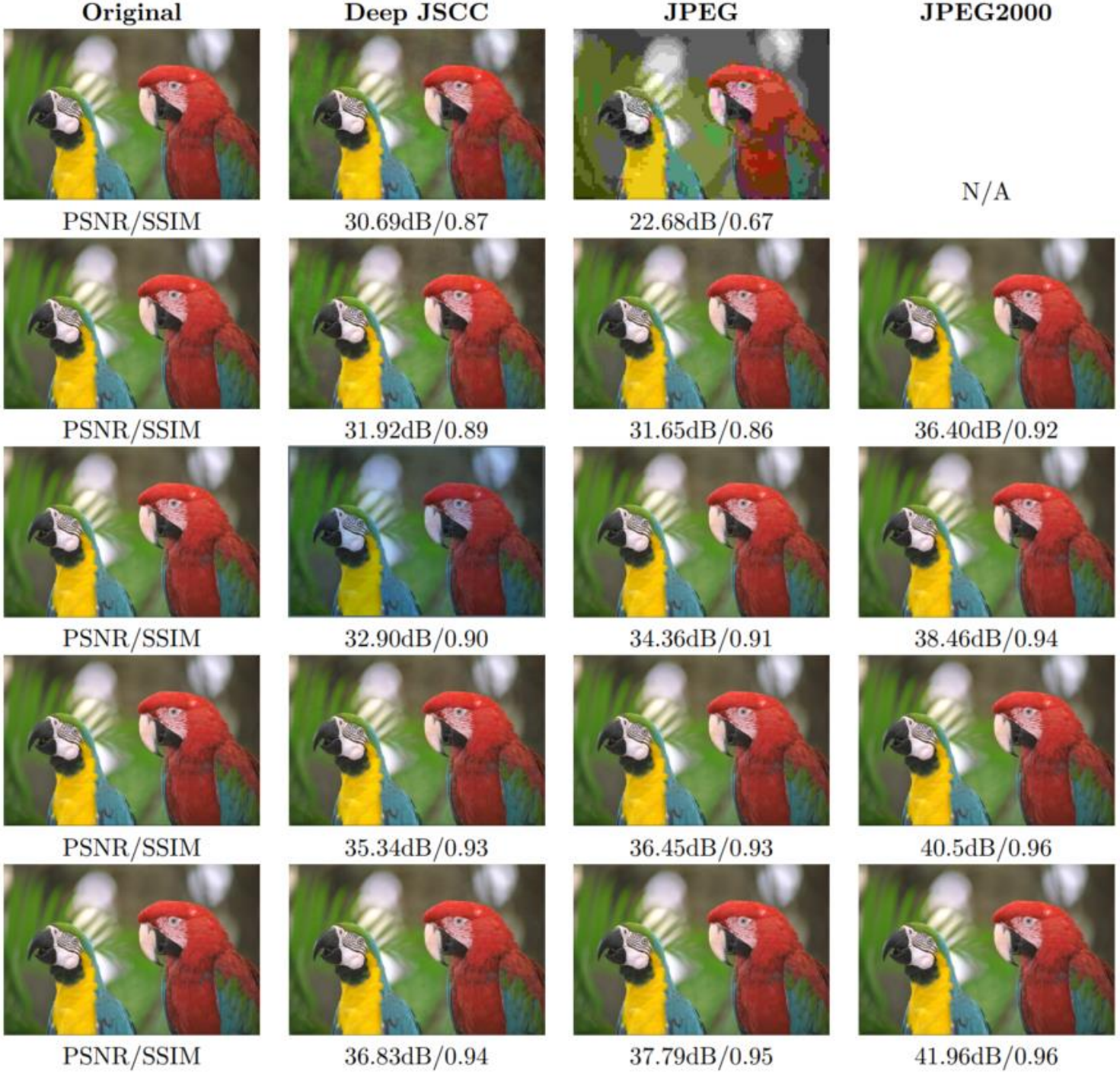
1dB,

4dB,

7dB,

13dB

19dB



SNRs

1dB,

4dB,

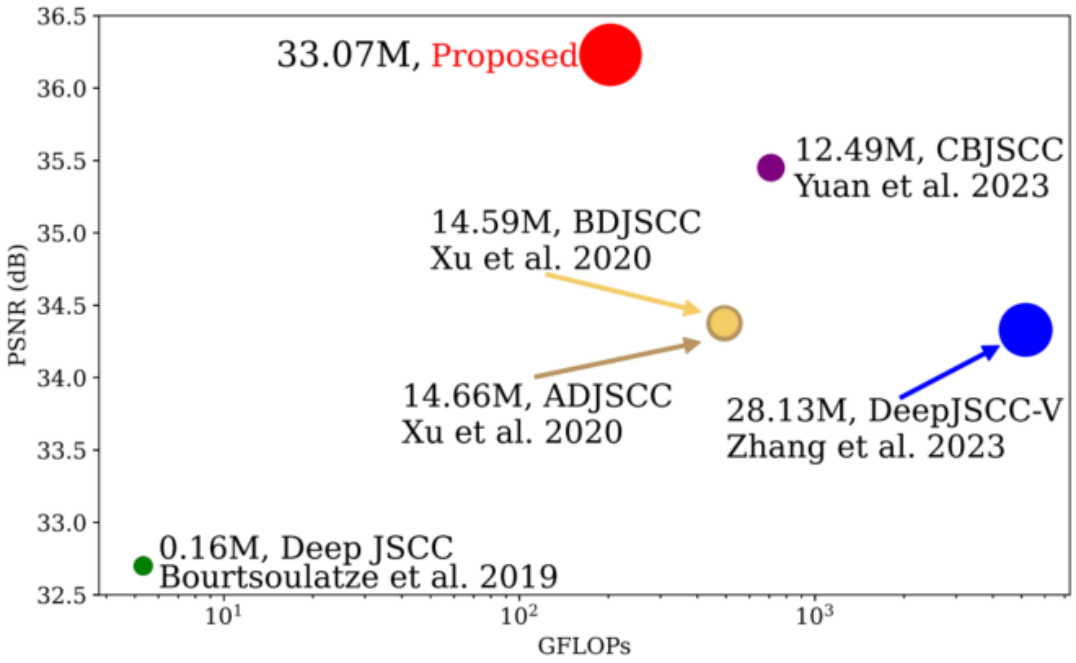
7dB,

13dB

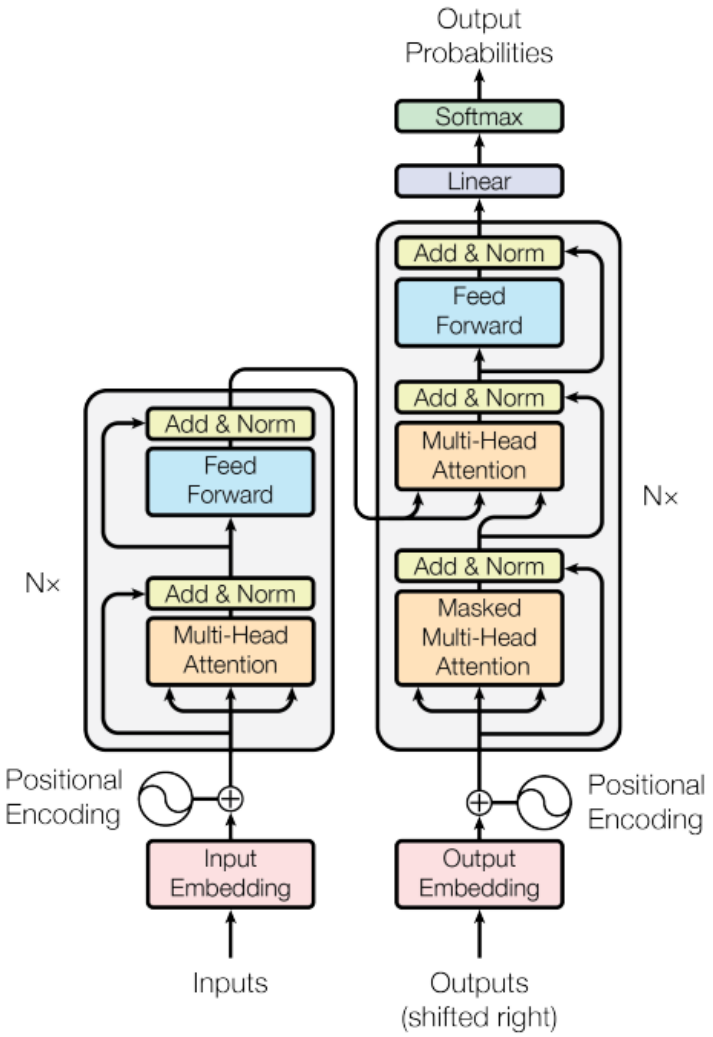
19dB

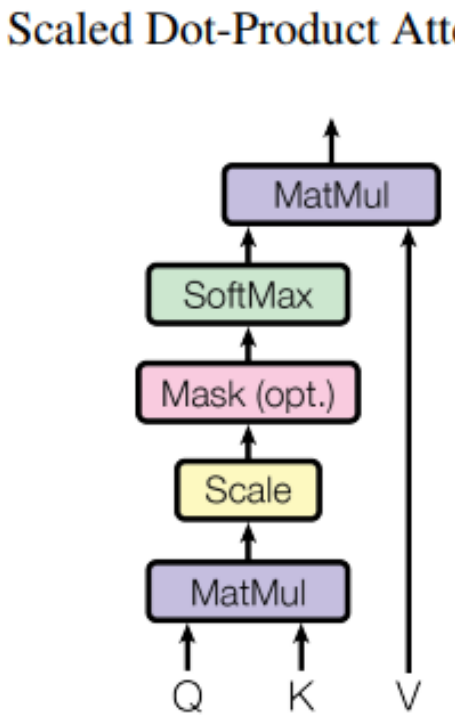
SwinJSCC

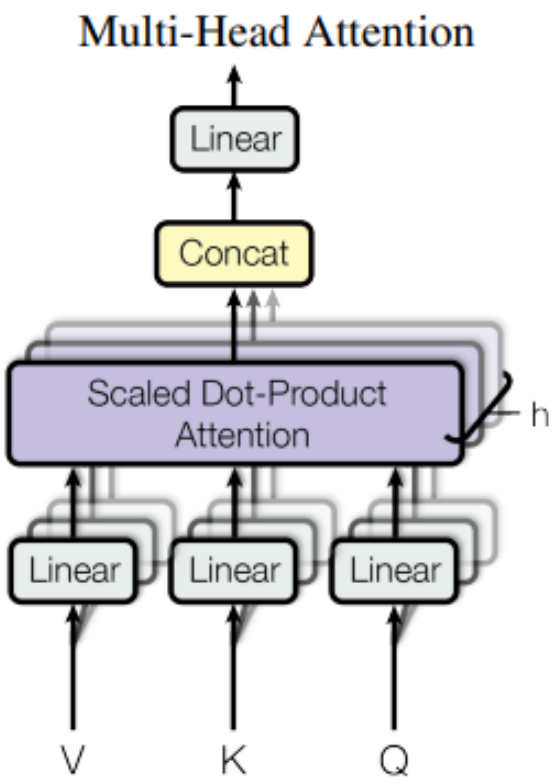
- Also proposed end-to-end JSCC using Swin Transformer called SwinJSCC
- Channel ModNet and Rate ModeNet
- SwinJSCC achieves better or comparable performance versus the state-of-the-art engineered BPG + 5G LDPC coded transmission system
- Code: https://github.com/semcomm/SwinJSCC



Yang, Ke, et al. "Swinjscc: Taming swin transformer for deep joint source-channel coding." *IEEE Transactions on Cognitive Communications and Networking* (2024).







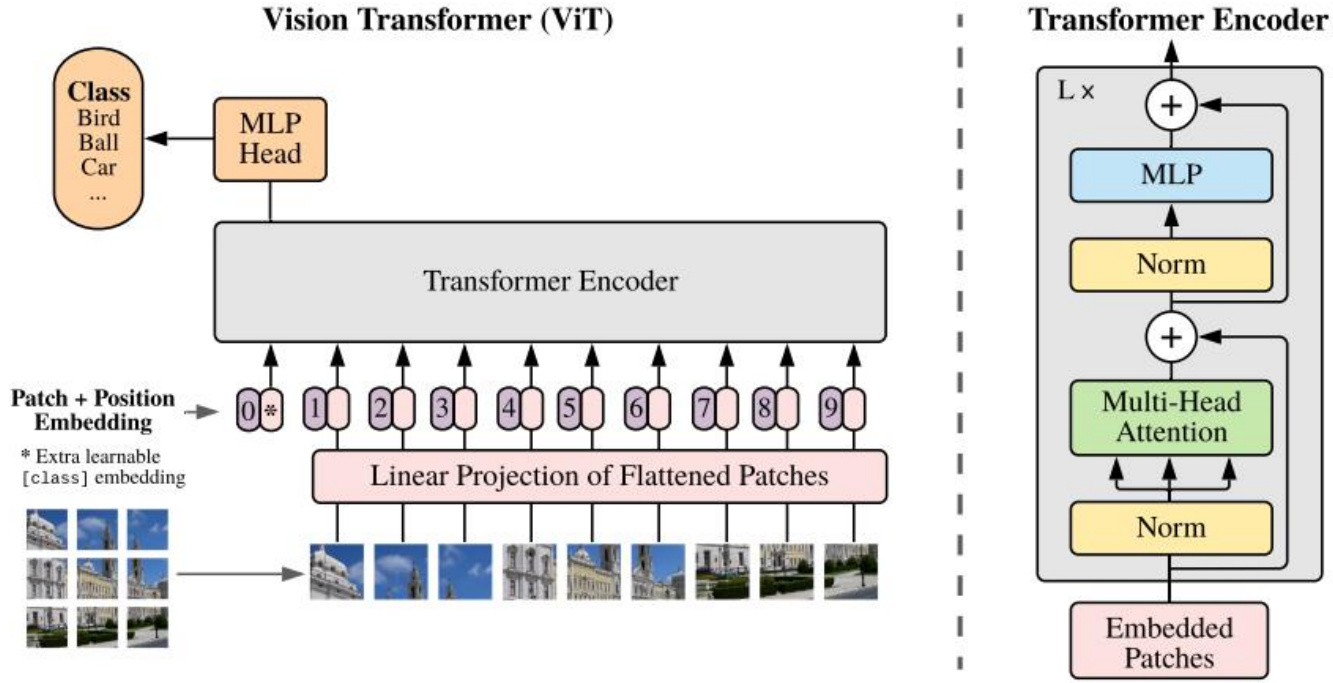
Multi-Head Attention consists of several attention layers running in parallel

Transformer: Encoder: N = 6 identical layers

Decoder: N = 6 identical layers

Vaswani, Ashish, et al. "Attention is all you need." *Advances in neural information processing systems* 30 (2017).

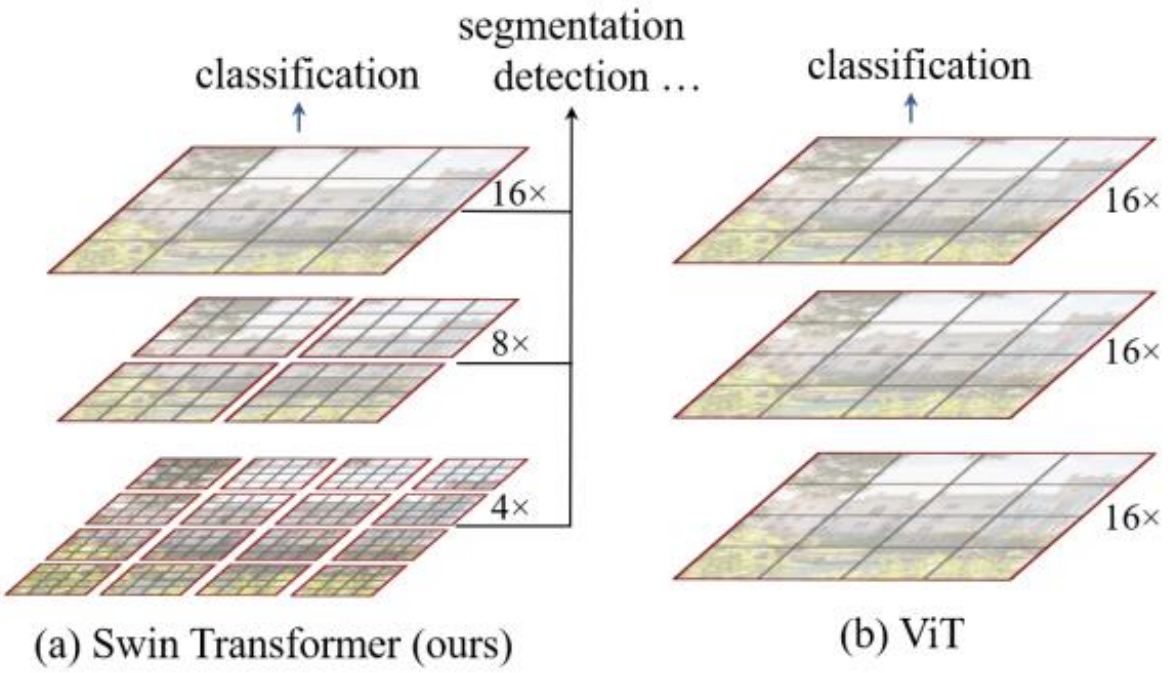
Vision Transformer



split an image into fixed-size patches, linearly embed each of them, add position embeddings, and feed the resulting sequence of vectors to a standard Transformer encoder

Dosovitskiy, Alexey, et al. "An image is worth 16x16 words: Transformers for image recognition at scale." *arXiv preprint arXiv:2010.11929* (2020).

Swin Transformer



(b) Vision Transformer:

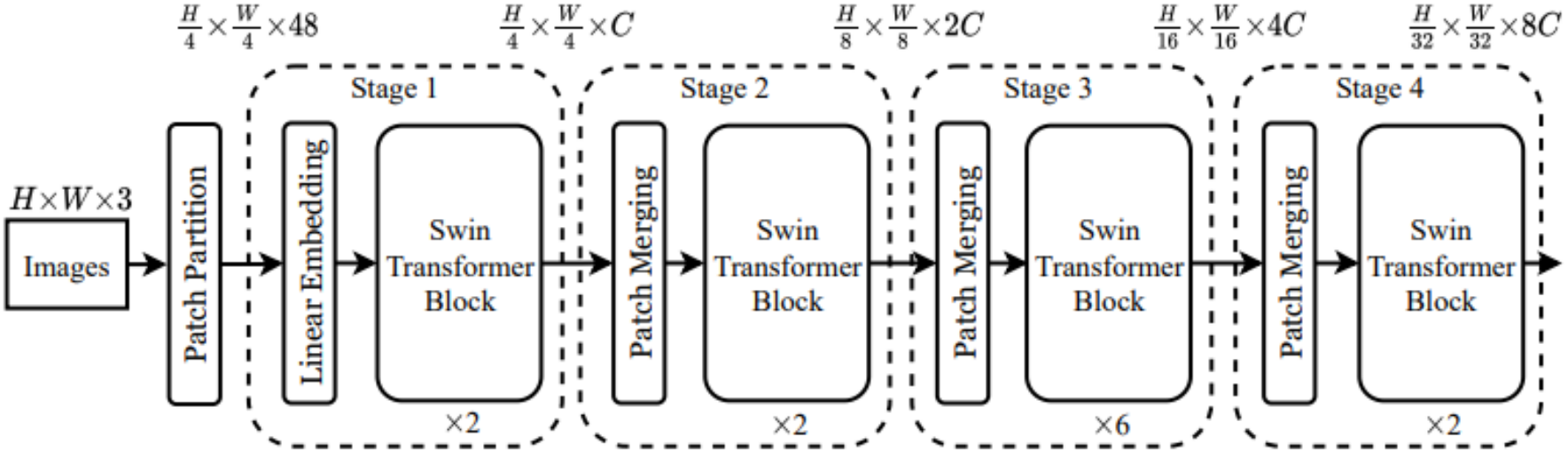
- single low resolution feature map
- quadratic computation complexity

(a) Swin Transformer:

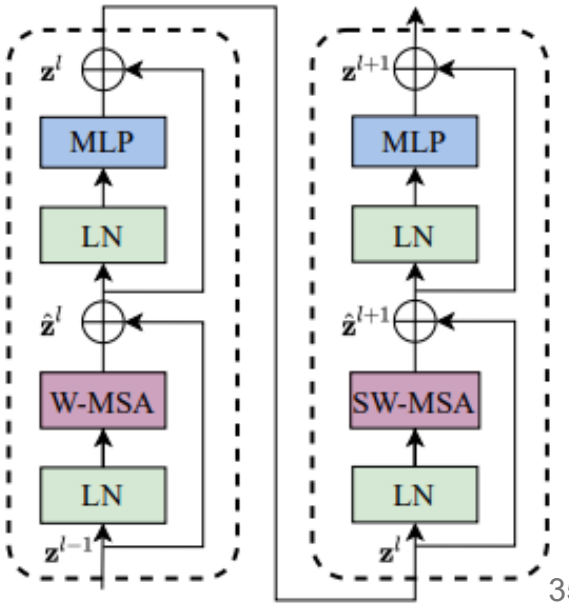
- hierarchical feature maps
- linear computation complexity

Liu, Ze, et al. "Swin transformer: Hierarchical vision transformer using shifted windows." *Proceedings of the IEEE/CVF international conference on computer vision*. 2021.

Swin Transformer (Liu et al. 21)

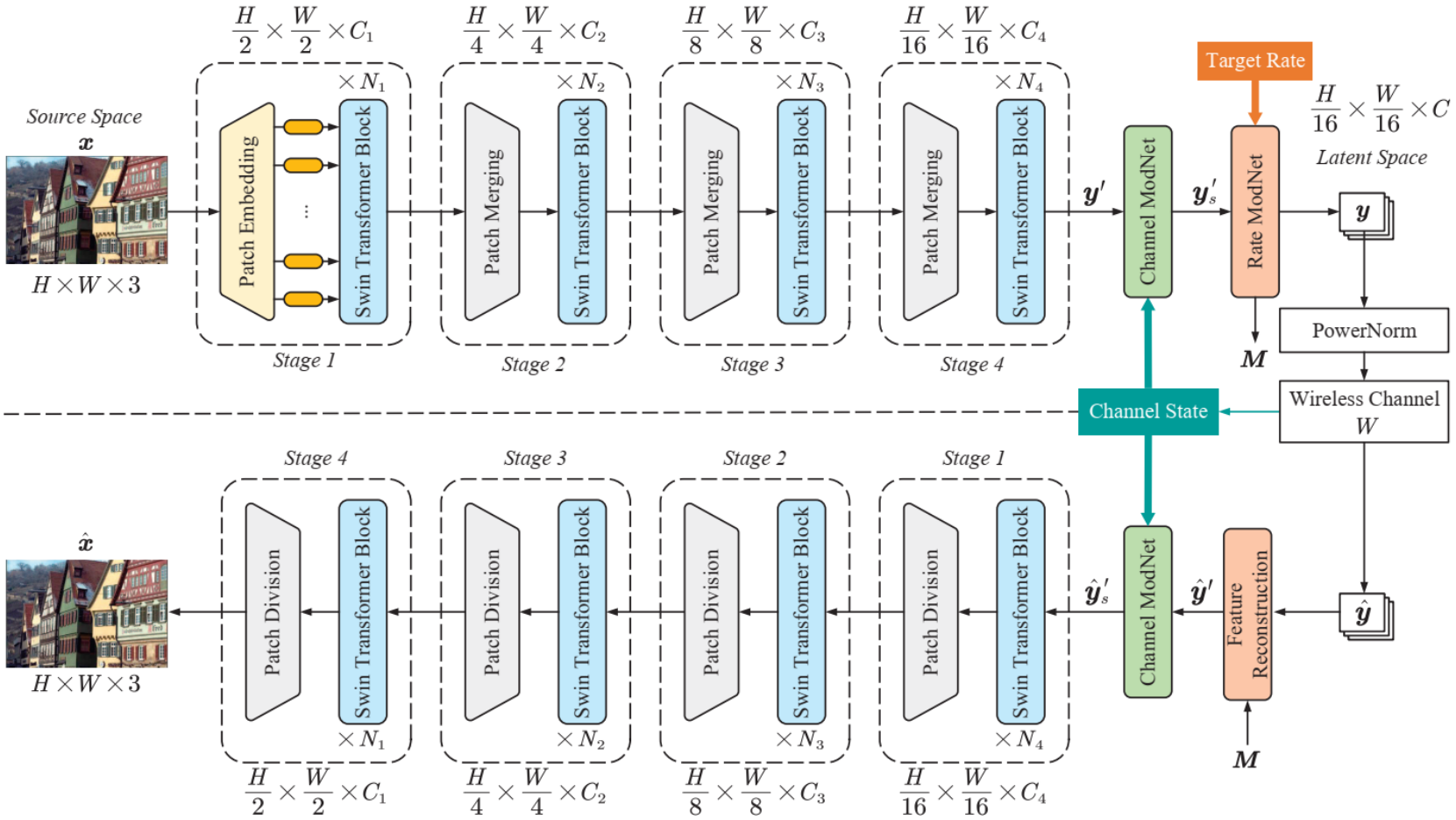


Swin Transformer architecture and two successive blocks

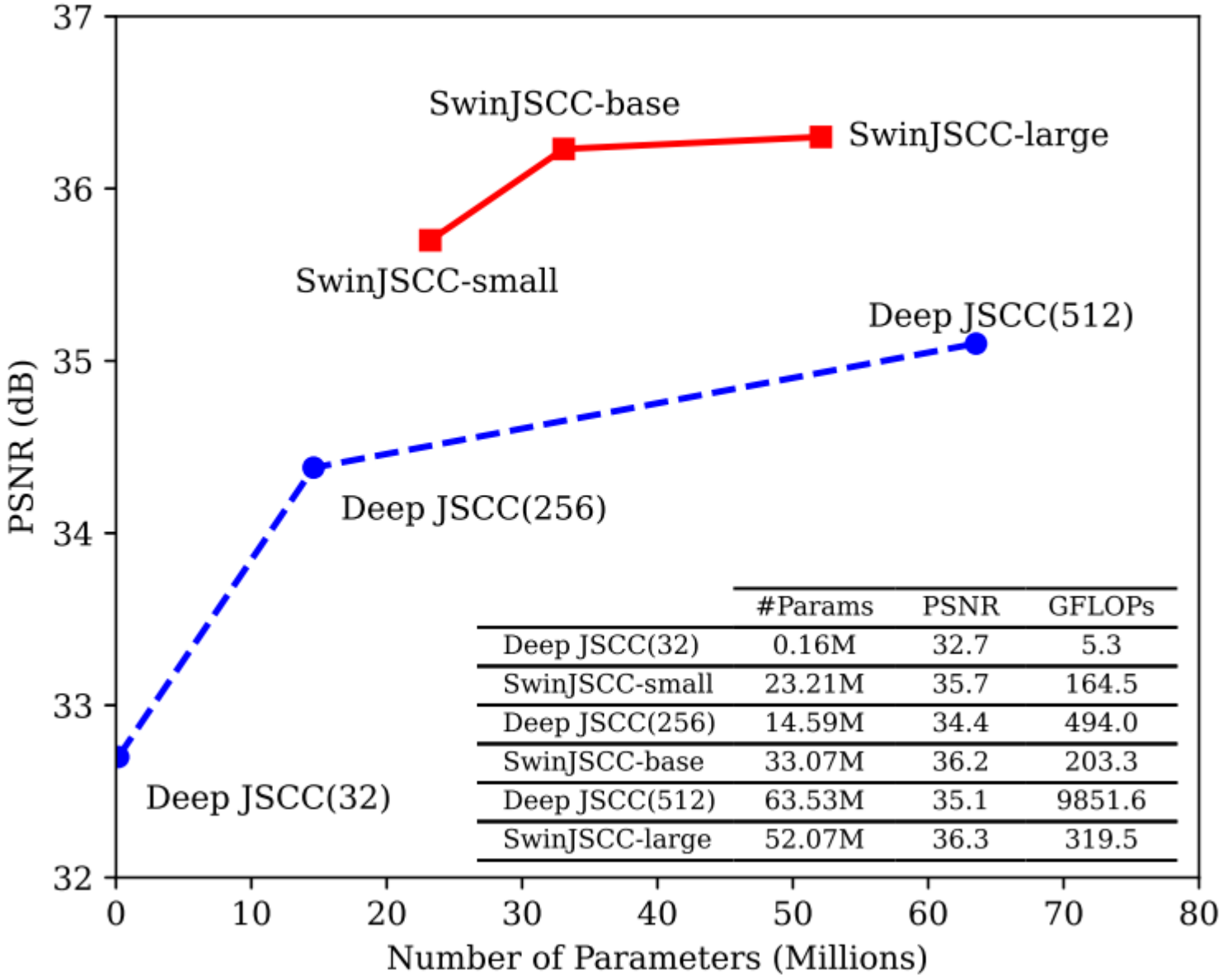


Liu, Ze, et al. "Swin transformer: Hierarchical vision transformer using shifted windows." *Proceedings of the IEEE/CVF international conference on computer vision*. 2021.

Architecture of SwinJSCC for wireless image transmission

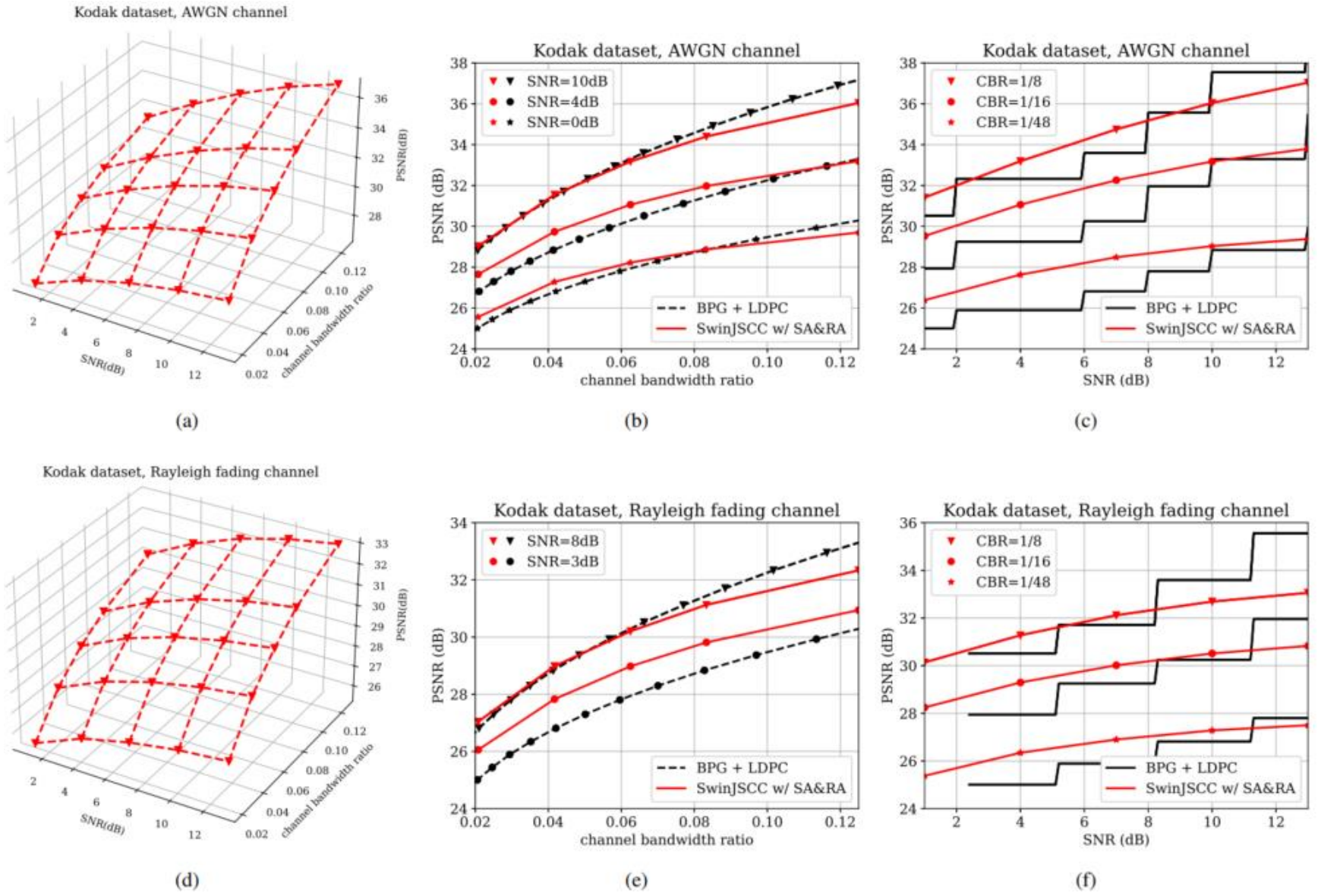


Yang, Ke, et al. "Swinjscc: Taming swin transformer for deep joint source-channel coding." *IEEE Transactions on Cognitive Communications and Networking* (2024).

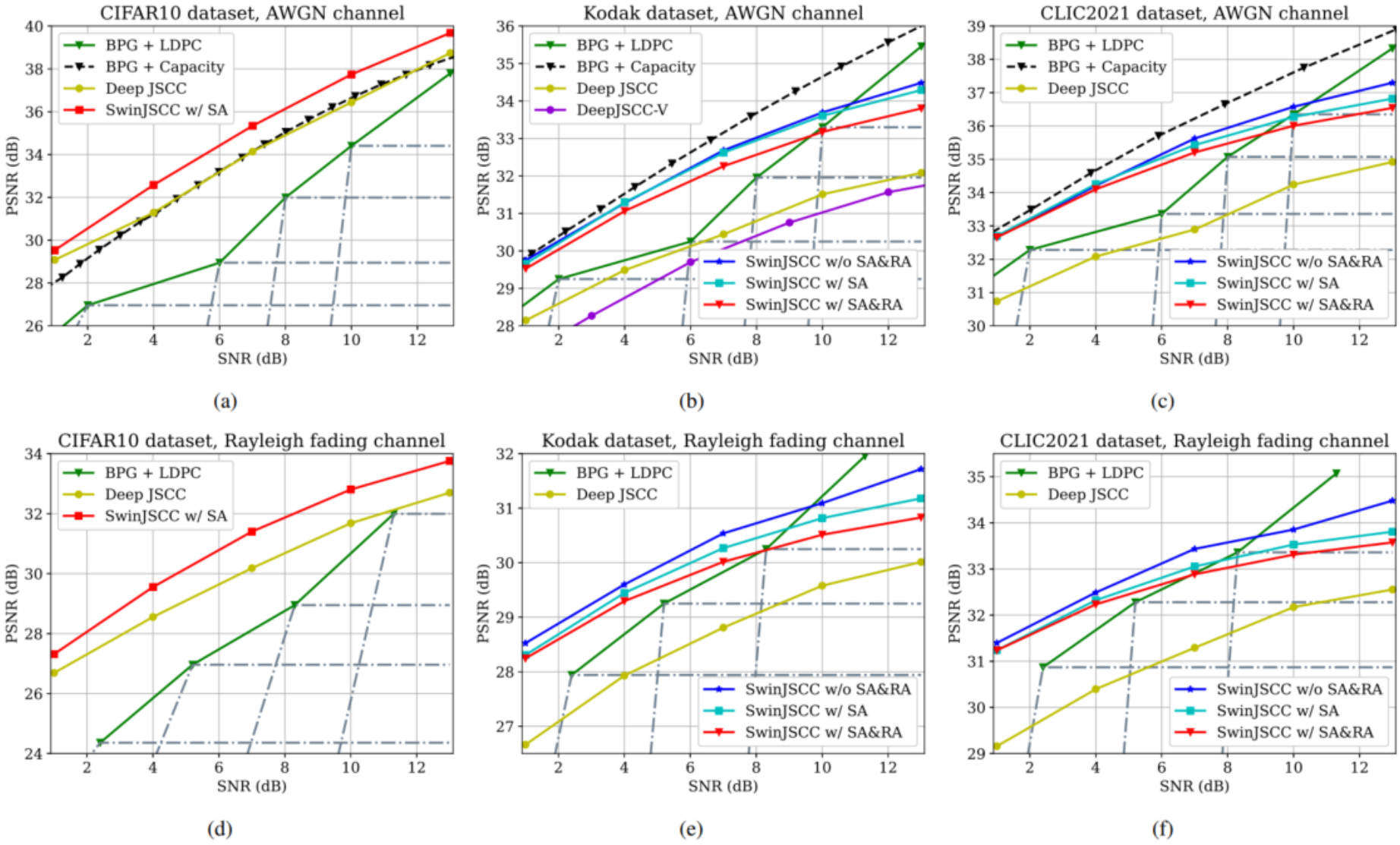


	#Params	PSNR	GFLOPs
Deep JSCC(32)	0.16M	32.7	5.3
SwinJSCC-small	23.21M	35.7	164.5
Deep JSCC(256)	14.59M	34.4	494.0
SwinJSCC-base	33.07M	36.2	203.3
Deep JSCC(512)	63.53M	35.1	9851.6
SwinJSCC-large	52.07M	36.3	319.5

Yang, Ke, et al. "Swinjscc: Taming swin transformer for deep joint source-channel coding." *IEEE Transactions on Cognitive Communications and Networking* (2024).



Yang, Ke, et al. "Swinjscc: Taming swin transformer for deep joint source-channel coding." *IEEE Transactions on Cognitive Communications and Networking* (2024).



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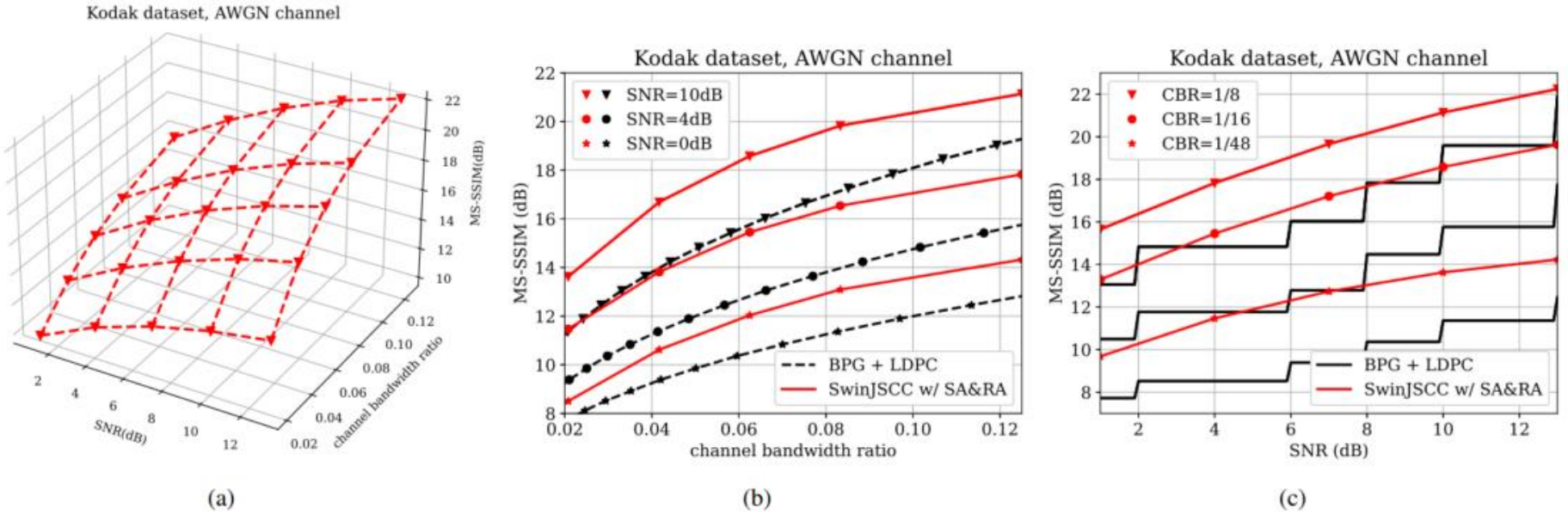


Fig. 12. SNR-rate-MS-SSIM comparison over AWGN and channel on Kodak dataset. (a) show the SNR-rate-MS-SSIM mesh obtained by our SwinJSCC w/ SA&RA model. (b) compares RD curves of different coded transmission schemes at SNR = 0dB, 4dB, and 10dB. (c) compares SNR-MS-SSIM curves under the CBR constraint CBR = 1/48, 1/16, 1/8.

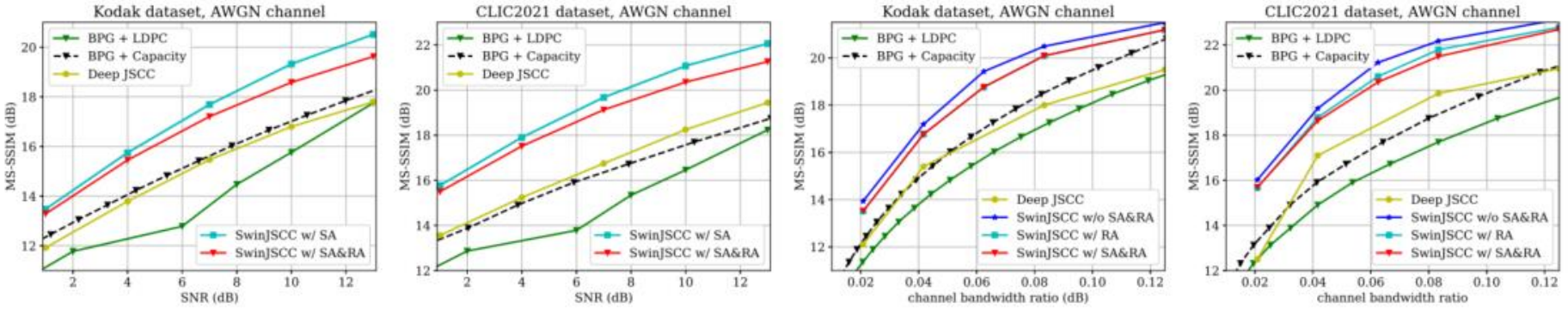


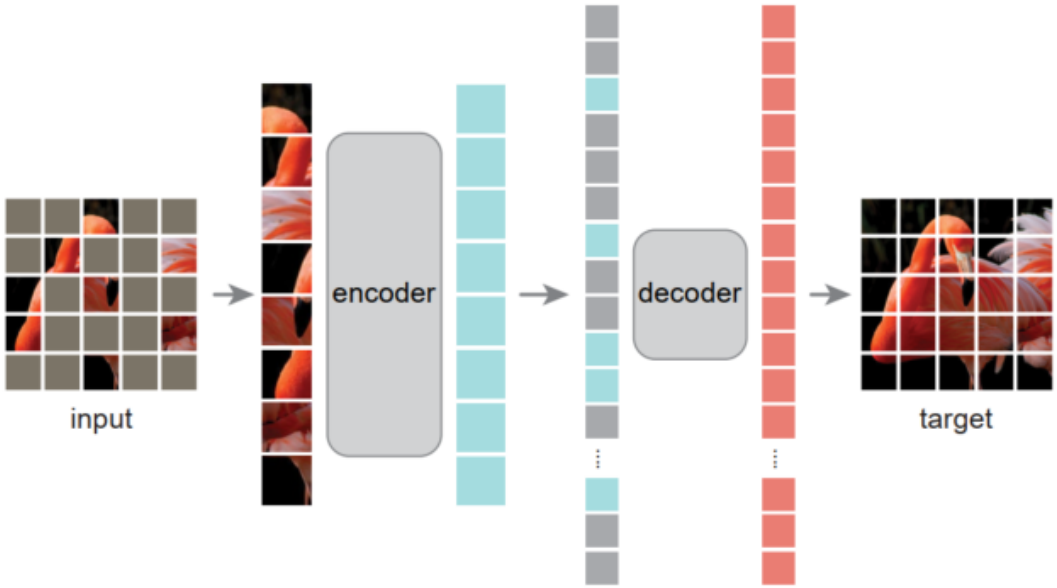
Fig. 13. (a)∼(b) MS-SSIM performance versus the SNR over the AWGN channel and the average CBR is set to 1/16. (c)∼(d) MS-SSIM performance versus the CBR over the AWGN channel at SNR = 10dB.

Yang, Ke, et al. "Swinjscc: Taming swin transformer for deep joint source-channel coding." *IEEE Transactions on Cognitive Communications and Networking* (2024).

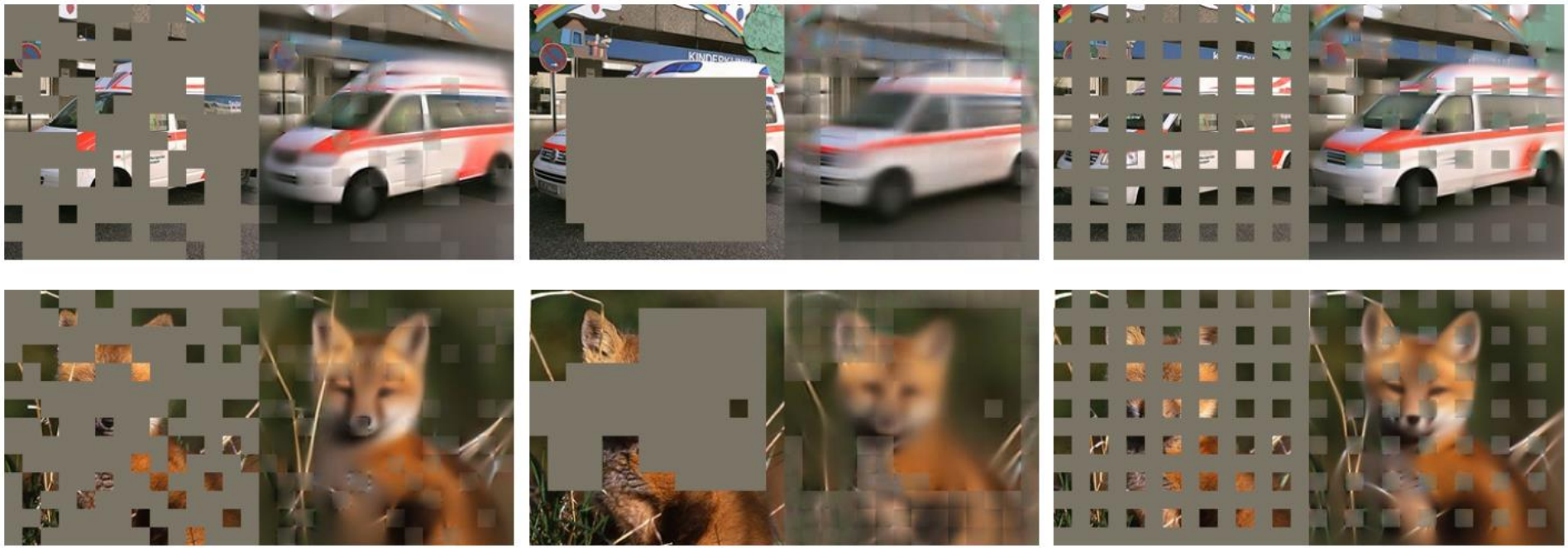
LoRA-MAEJSCC: Masked Autoencoders for Semantic Communications using Low-Rank Adaptation

Masked autoencoder (MAE)

- Asymmetric encoder-decoder architecture:
  - **encoder** that operates only on the visible subset of patches (without mask tokens)
  - **lightweight decoder** that reconstructs the original image from the latent representation and mask tokens.
- Random masking, 75% masking ratio
- Loss function: mean squared error (MSE) between the reconstructed and original images in the pixel space



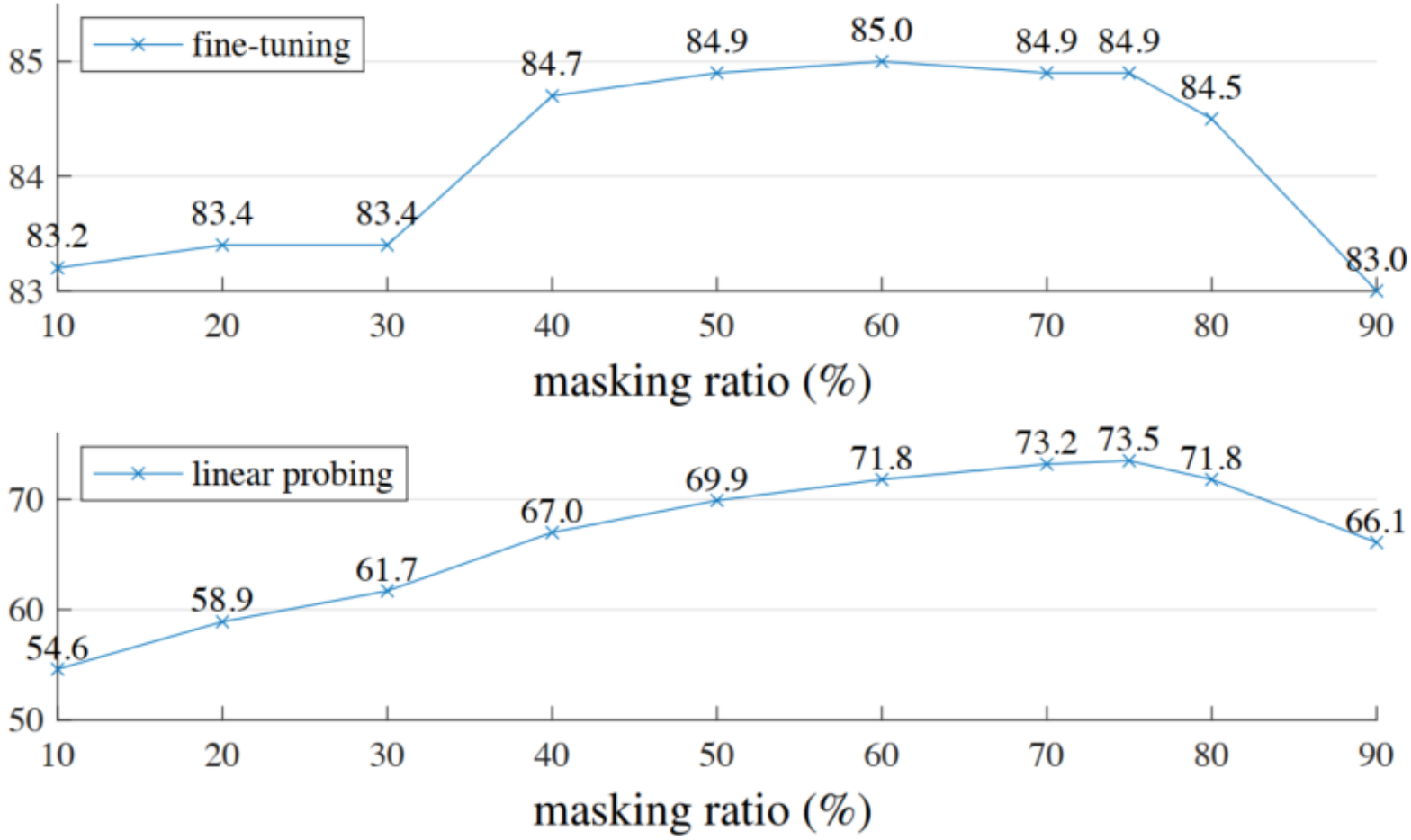
He, Kaiming, et al. "Masked autoencoders are scalable vision learners." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022.



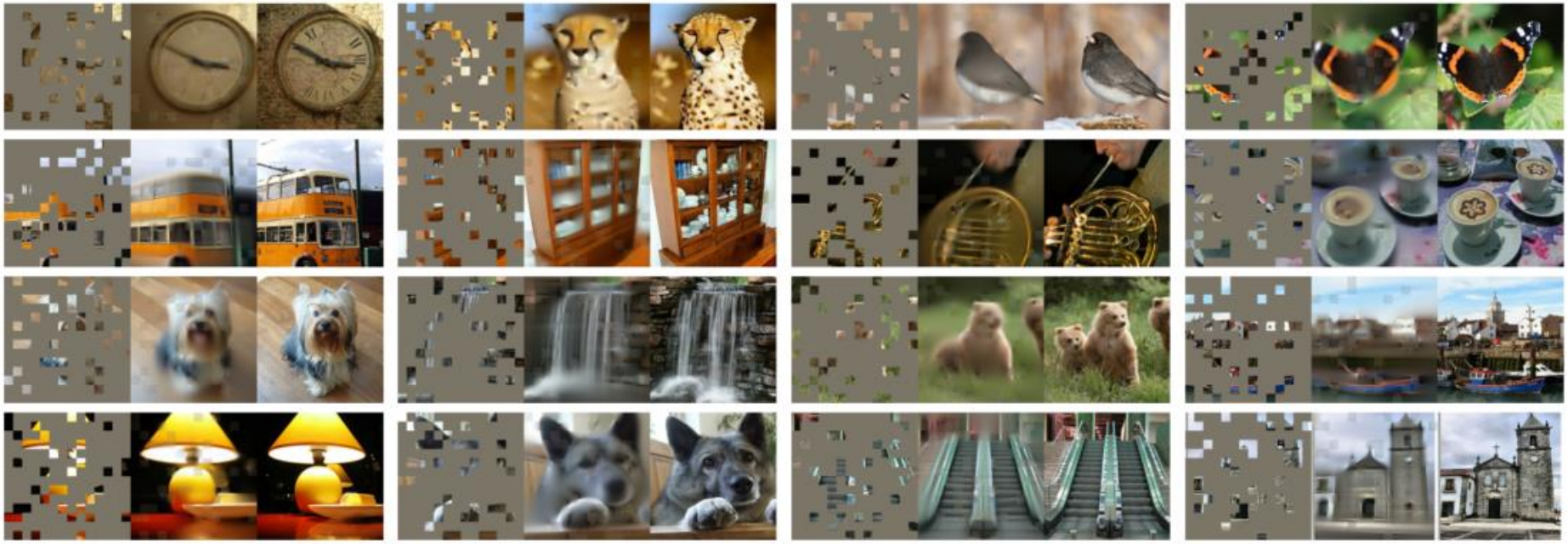
random 75% block 50% grid 75%

He, Kaiming, et al. "Masked autoencoders are scalable vision learners." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022.

MAE – Masking ratio



A high masking ratio (75%) works well for both fine-tuning (top) and linear probing (bottom).
(The y-axes are ImageNet-1K validation accuracy (%))



ImageNet, 80% masking (39 out of 196). The masked image (left), MAE reconstruction (middle), and the ground-truth (right)



CoCo, 80% masking (39 out of 196)

He, Kaiming, et al. "Masked autoencoders are scalable vision learners." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022.

- Decoder depths and widths:

blocks	ft	lin
1	84.8	65.5
2	**84.9**	70.0
4	**84.9**	71.9
8	**84.9**	**73.5**
12	84.4	73.3

dim	ft	lin
128	**84.9**	69.1
256	84.8	71.3
512	**84.9**	**73.5**
768	84.4	73.1
1024	84.3	73.1

- Wall-clock time: (ViT-L 24 blocks, 1024-d)

encoder	dec. depth	ft acc	hours	speedup
ViT-L, w/ [M]	8	84.2	42.4	-
ViT-L	8	84.9	15.4	2.8×
ViT-L	1	84.8	11.6	**3.7**×
ViT-H, w/ [M]	8	-	119.6†	-
ViT-H	8	85.8	34.5	3.5×
ViT-H	1	85.9	29.3	**4.1**×

He, Kaiming, et al. "Masked autoencoders are scalable vision learners." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022.

LoRA-MAEJSCC: Masked Autoencoders for Semantic Communications using Low-Rank Adaptation

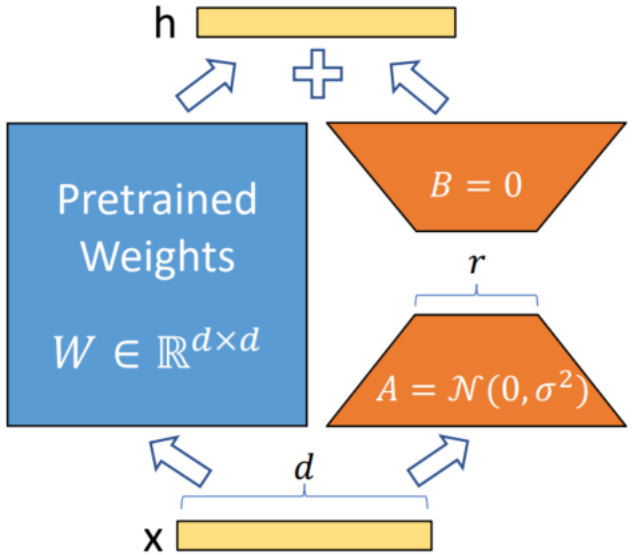
Low-Rank Adaptation (LoRA):

$$W' = W + \Delta W, \Delta W = BA$$

- $A \in R^{r \times d}, B \in R^{d \times r}$
- Rank $r \ll d$

Benefits:

- Reduce trainable parameters by 100–10,000× ➔ lower GPU memory usage, faster training
- Easy to switch between multiple task-specific adapters

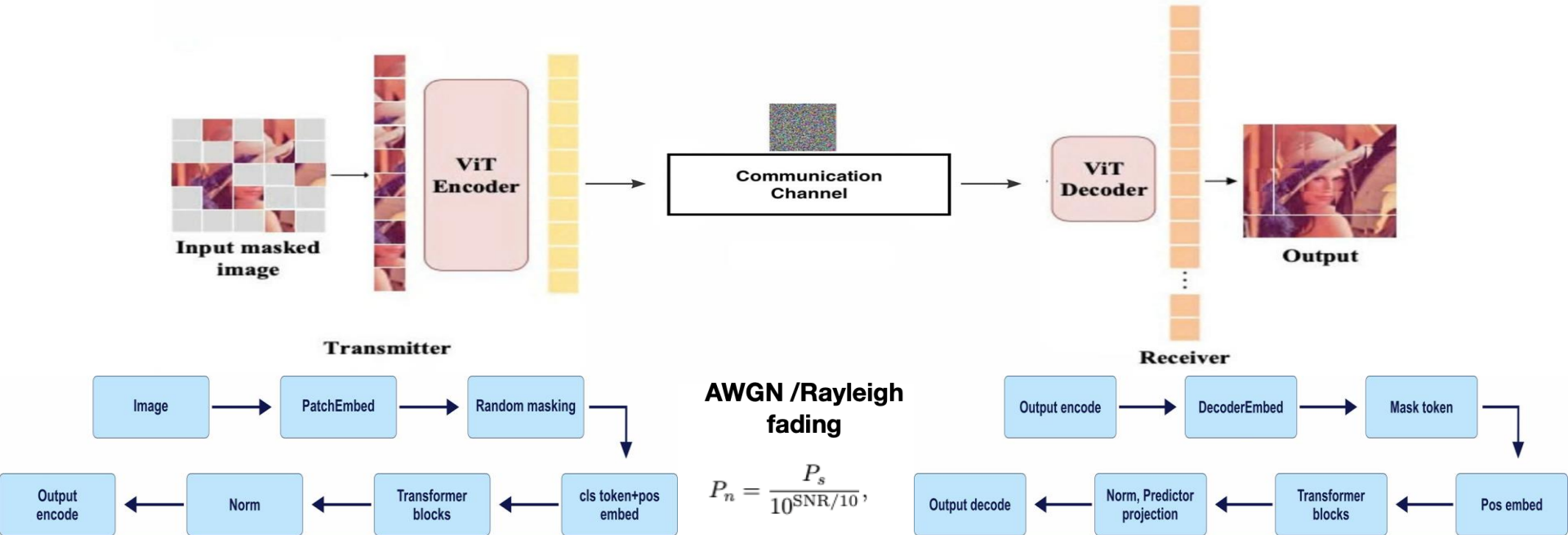


Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, and Weizhu Chen. "Lora: Low-rank adaptation of large language models." *Iclr* 1, no. 2 (2022): 3.

LoRA

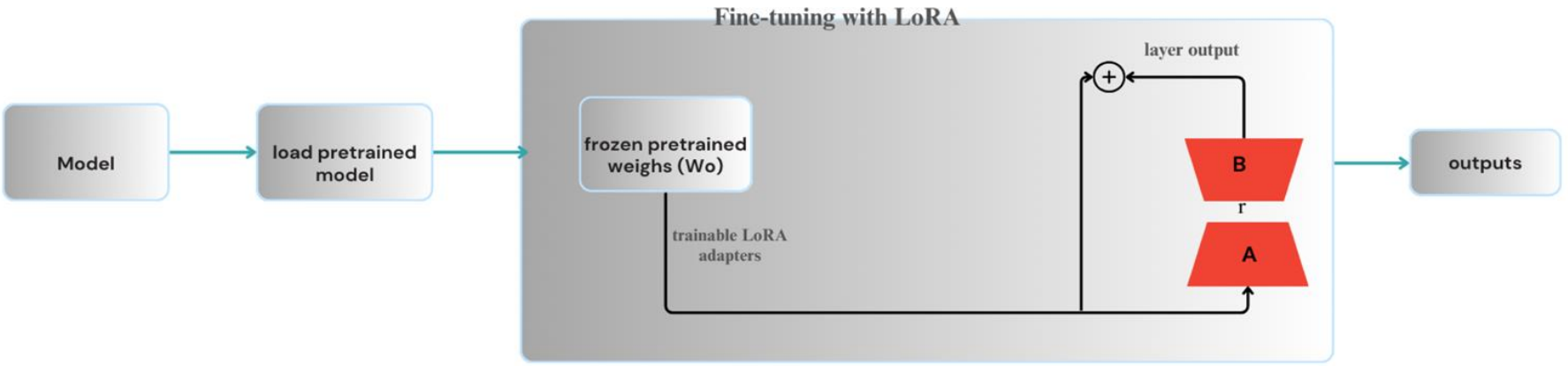
Method	# of Trainable Parameters	WikiSQL Accuracy (%)	MNLI-m Accuracy (%)	SAMSum R1/R2/RL
GPT-3 175B (Fine-Tune)	175,255.8M	73.0	89.5	52.0/28.0/44.5
GPT-3 175B (Bias Only)	14.2M	71.3	91.0	51.3/27.4/43.5
GPT-3 175B (PrefixEmbed)	3.2M	63.1	88.6	48.3/24.2/40.5
GPT-3 175B (PrefixLayer)	20.2M	70.1	89.5	50.8/27.3/43.5
GPT-3 175B (LoRA)	4.7M	73.4	91.3	52.1/28.3/44.3
GPT-3 175B (LoRA)	37.7M	**73.8**	**91.7**	**53.2/29.2/45.0**

LoRA-MAEJSCC: Masked Autoencoders for Semantic Communications using Low-Rank Adaptation

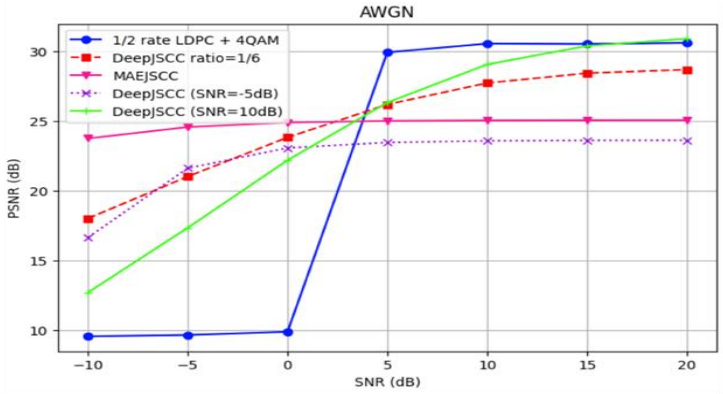


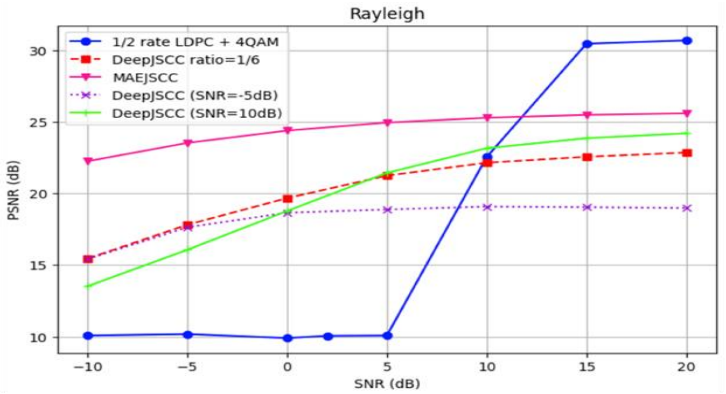
Nguyen, Thuy An, Tran Ngoc Thinh, Toan Van Nguyen, and Phuong Luu Vo. "LoRA-MAEJSCC: Masked Autoencoders for Semantic Communications using Low-Rank Adaptation." In *2025 International Conference on Advanced Technologies for Communications (ATC)*, pp. 1-6. IEEE, 2025.

LoRA-MAEJSCC: Masked Autoencoders for Semantic Communications using Low-Rank Adaptation



Experiments





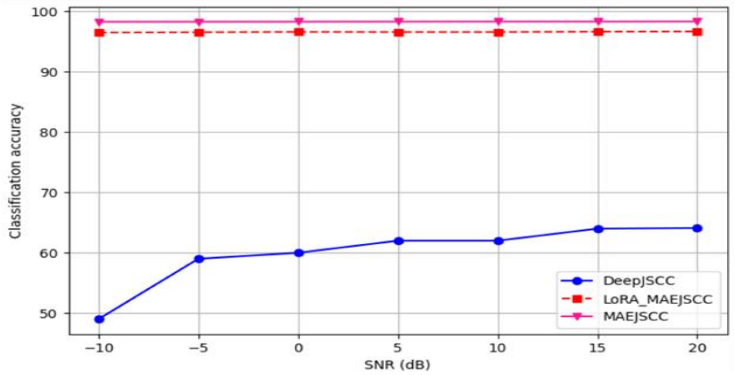
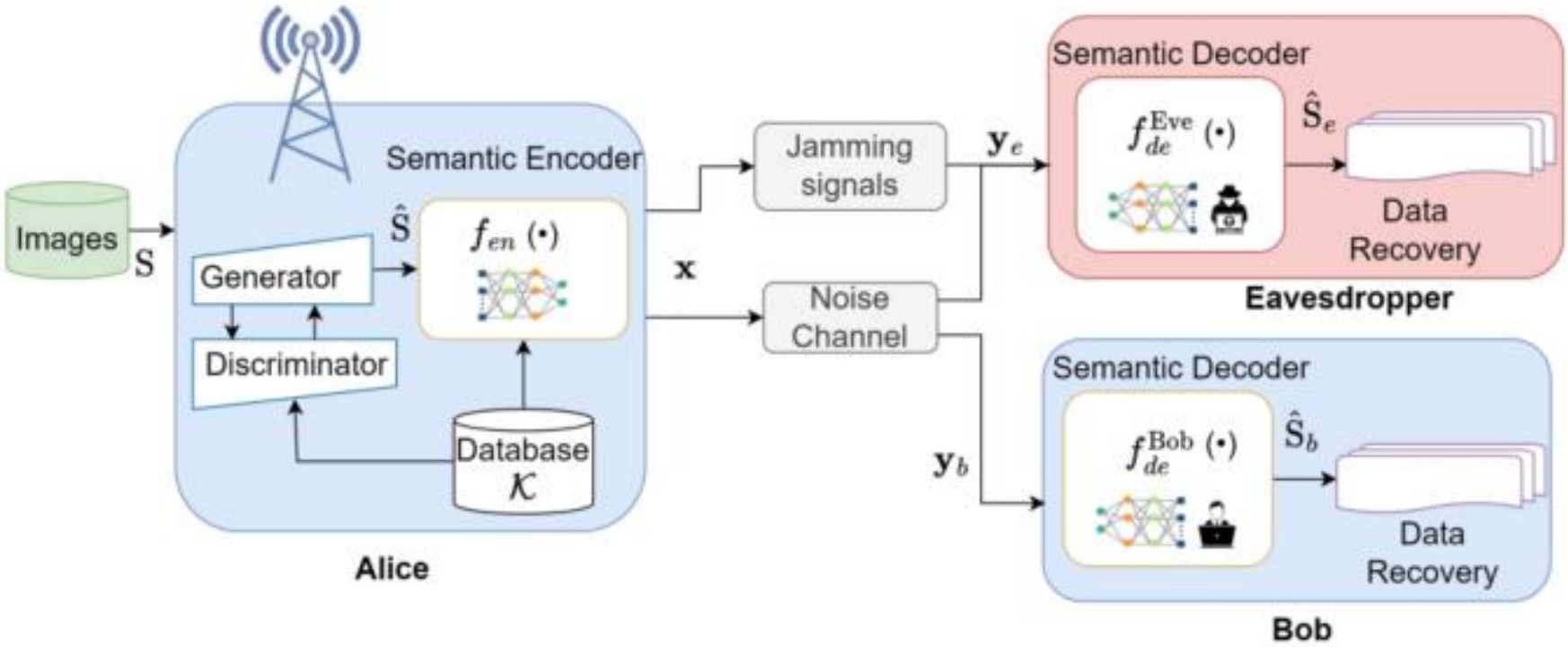


TABLE I: Comparison of classification accuracy, trainable parameters, and checkpoint size between MAEJSCC and LoRA-MAEJSCC on CIFAR-10.

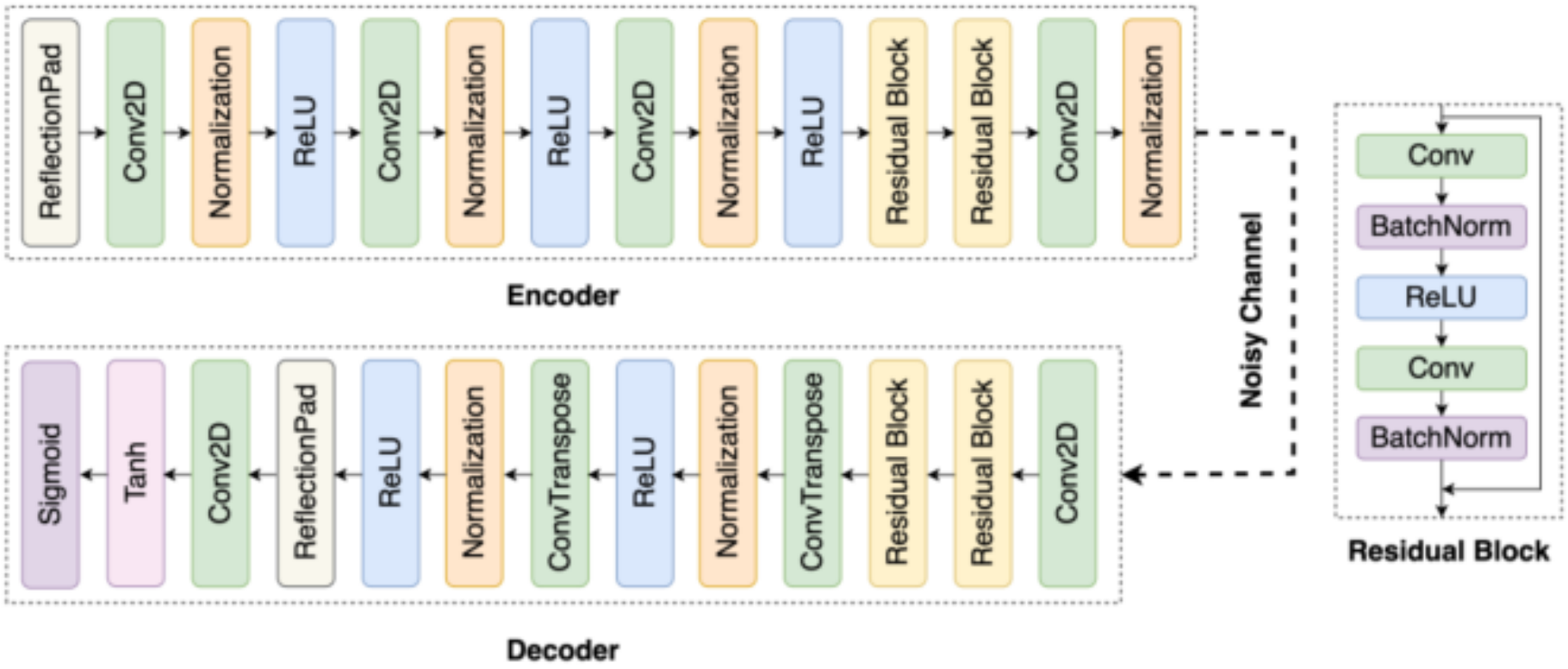
Model	Acc. (%)	Params	Ckpt (MB)
MAEJSCC (Full)	97.9	86.7M	982.0
LoRA-MAEJSCC	96.6	0.91M	337.9

Security Improvement for Deep Learning-based Semantic Communication Systems

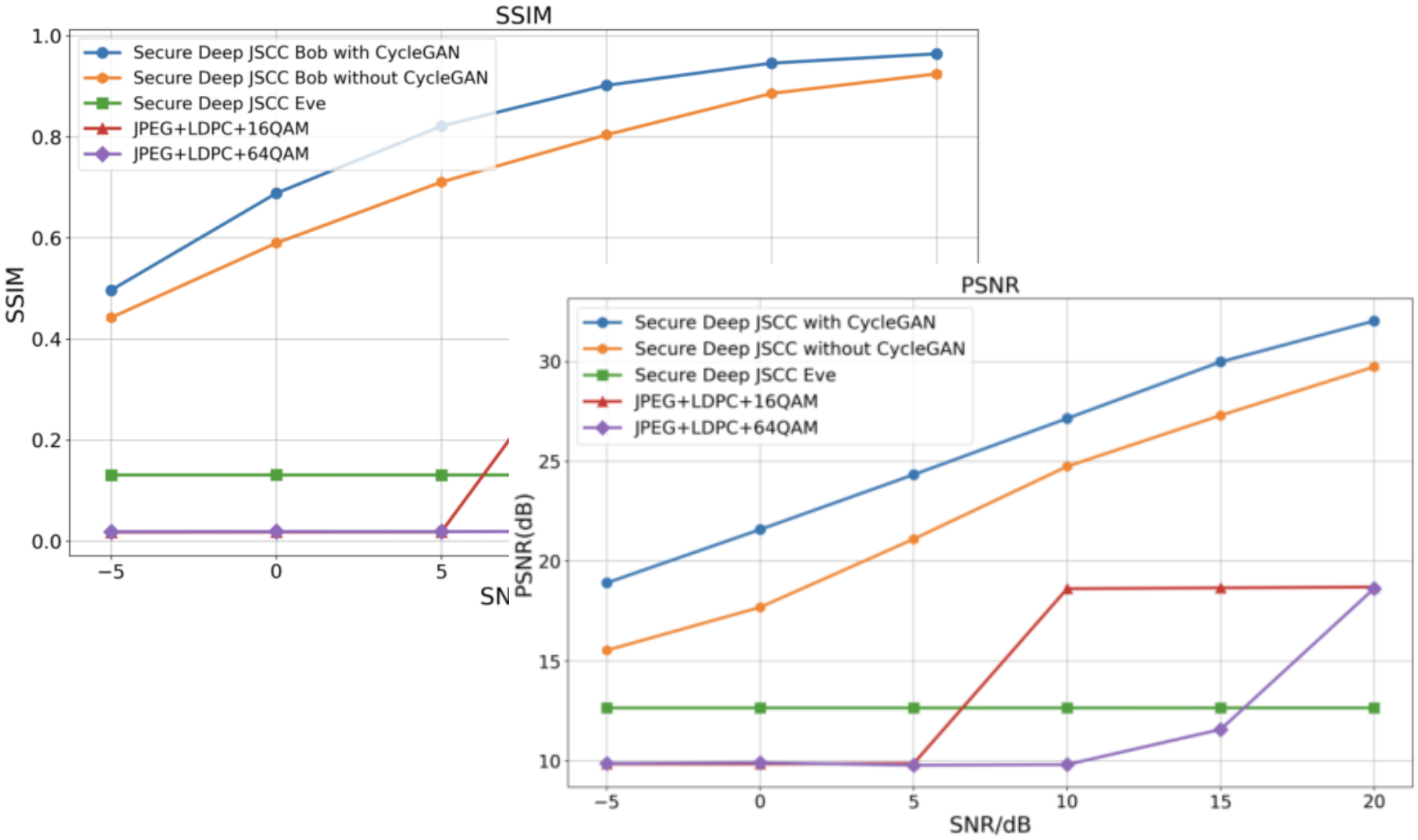


Hung Ngo Luu Tan, Van-Dinh Nguyen, Thien Huynh-The, Toan-Van Nguyen, and Phuong Luu Vo. "Security improvement for deep learning-based semantic communication systems." In 2024 International Conference on Advanced Technologies for Communications (ATC), pp. 717-721. IEEE, 2024.

Security Improvement for Deep Learning-based Semantic Communication Systems – Deep JSCC structure



Security Improvement for Deep Learning-based Semantic Communication Systems - Performance evaluation



Security Improvement for Deep Learning-based Semantic Communication Systems – Security evaluation



(a) Original image. (b) Deep JSCC image.

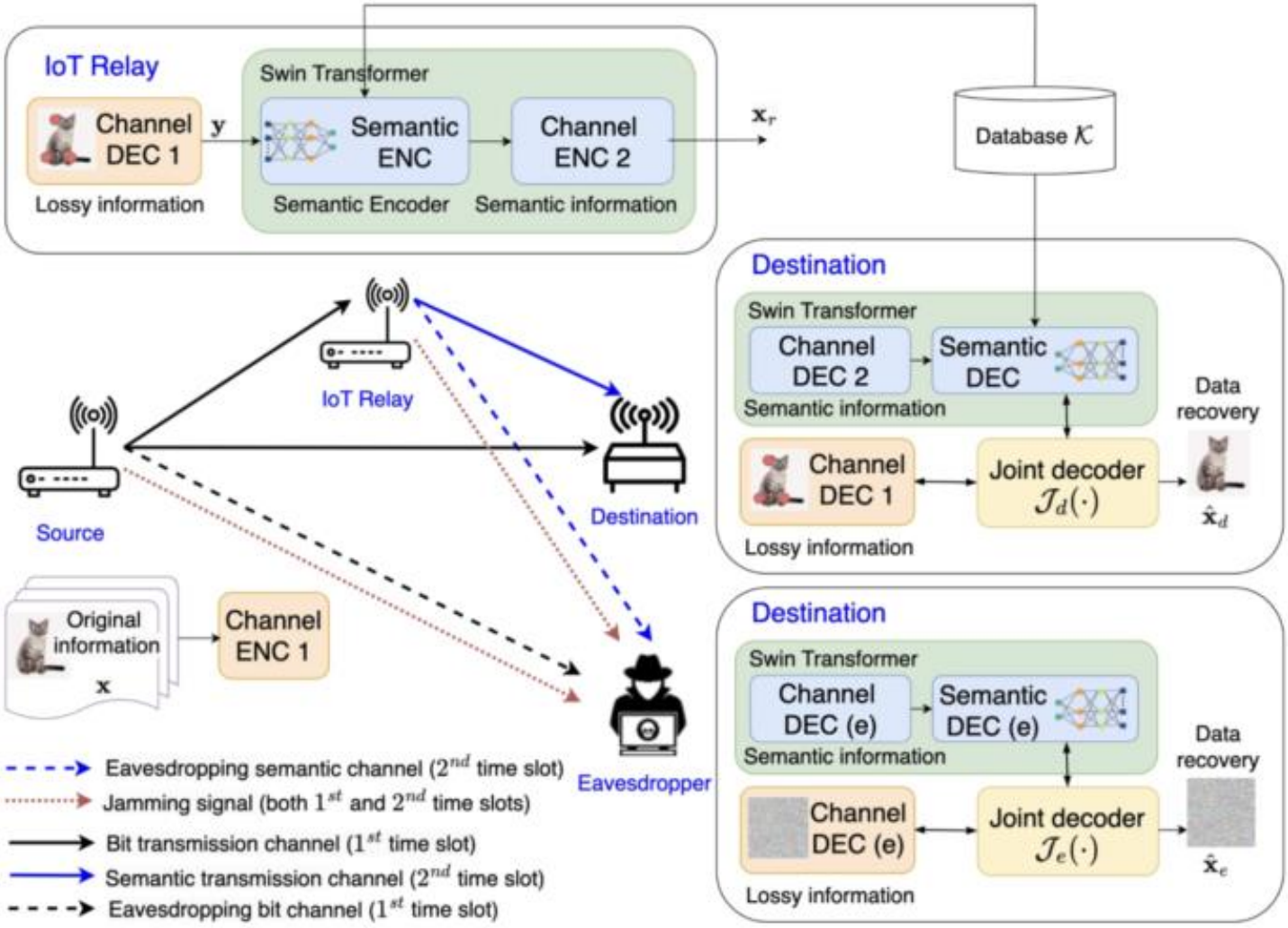
Fig. 3: Eve's reconstructed image at SNR $= 0$ dB.



(a) Original (b) Bob (c) Eve

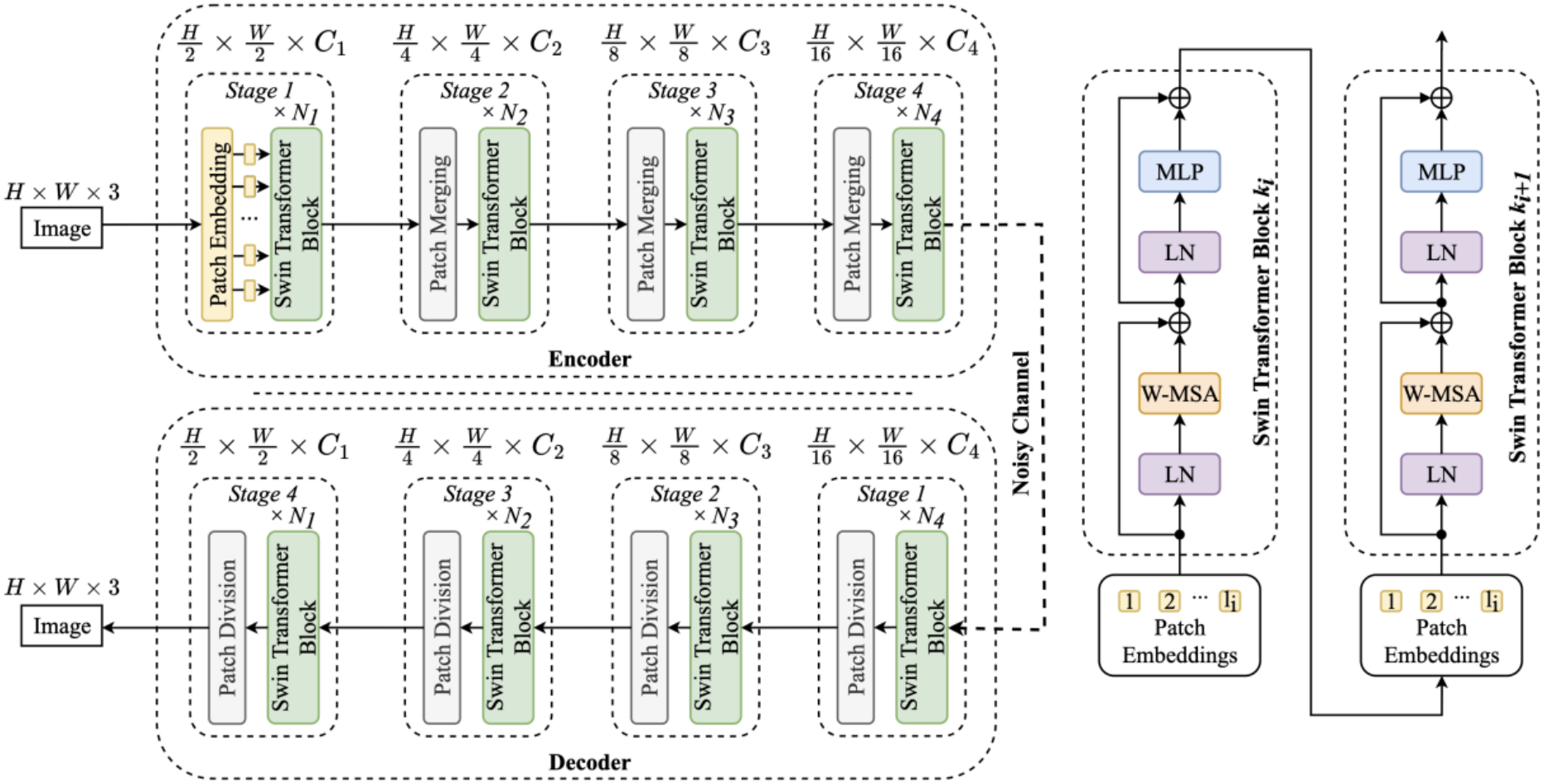
Fig. 6: Visual results from models trained on SNR$=$ 15 dB and tested on the same channel.

Secure Cooperative Semantic Communication with Swin Transformer and Jamming



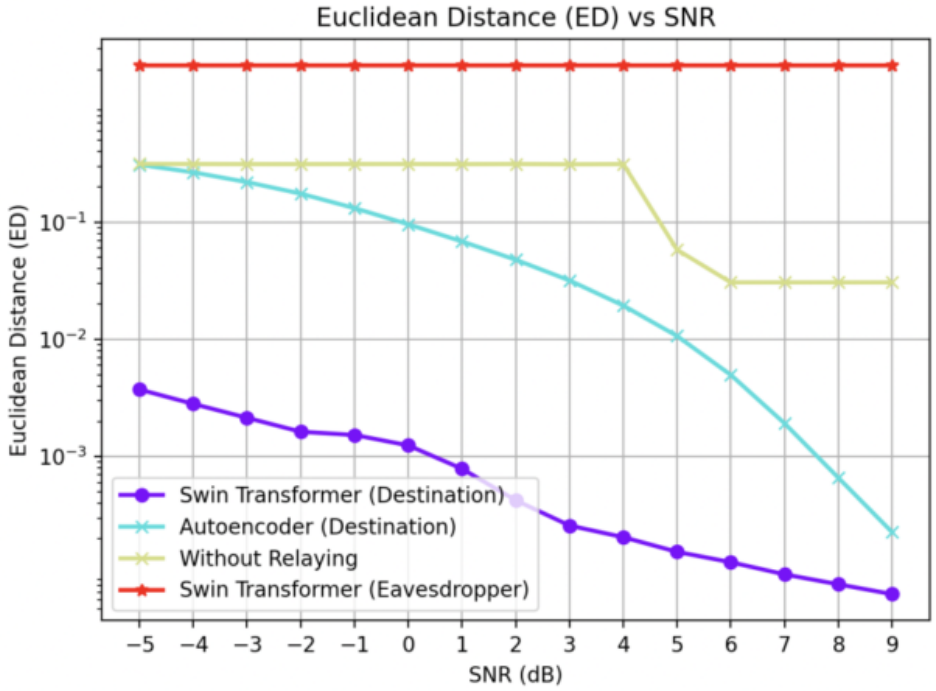
Tan, Hung Ngo Luu, Van Toan Nguyen, and Phuong Luu Vo. "Secure Cooperative Semantic Communication with Swin Transformer and Jamming." In ICCCI 2025.

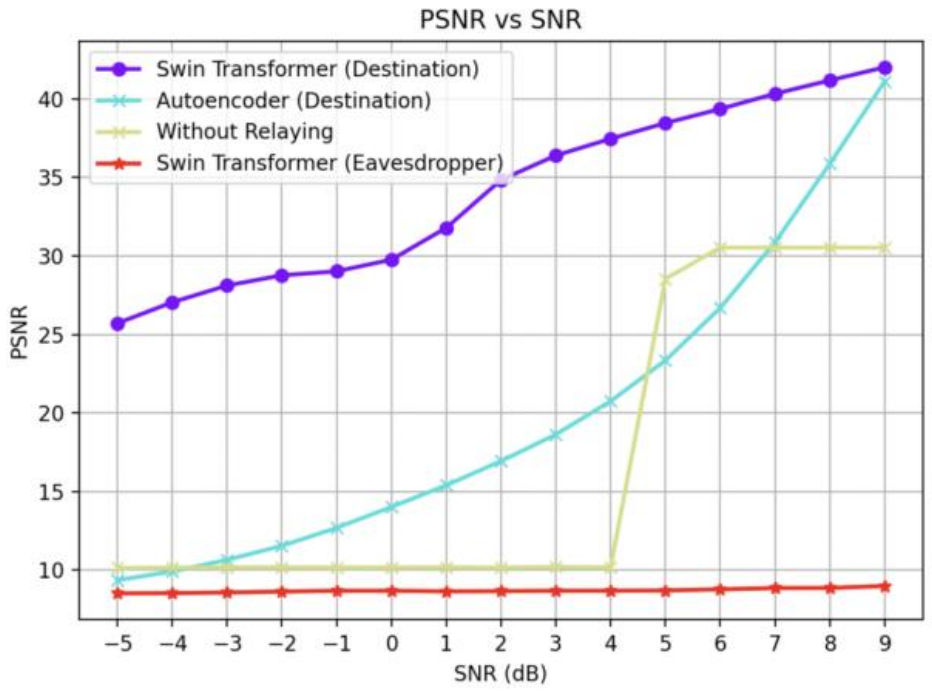
Secure Cooperative Semantic Communication with Swin Transformer and Jamming – Swin Transformer encoder and decoder

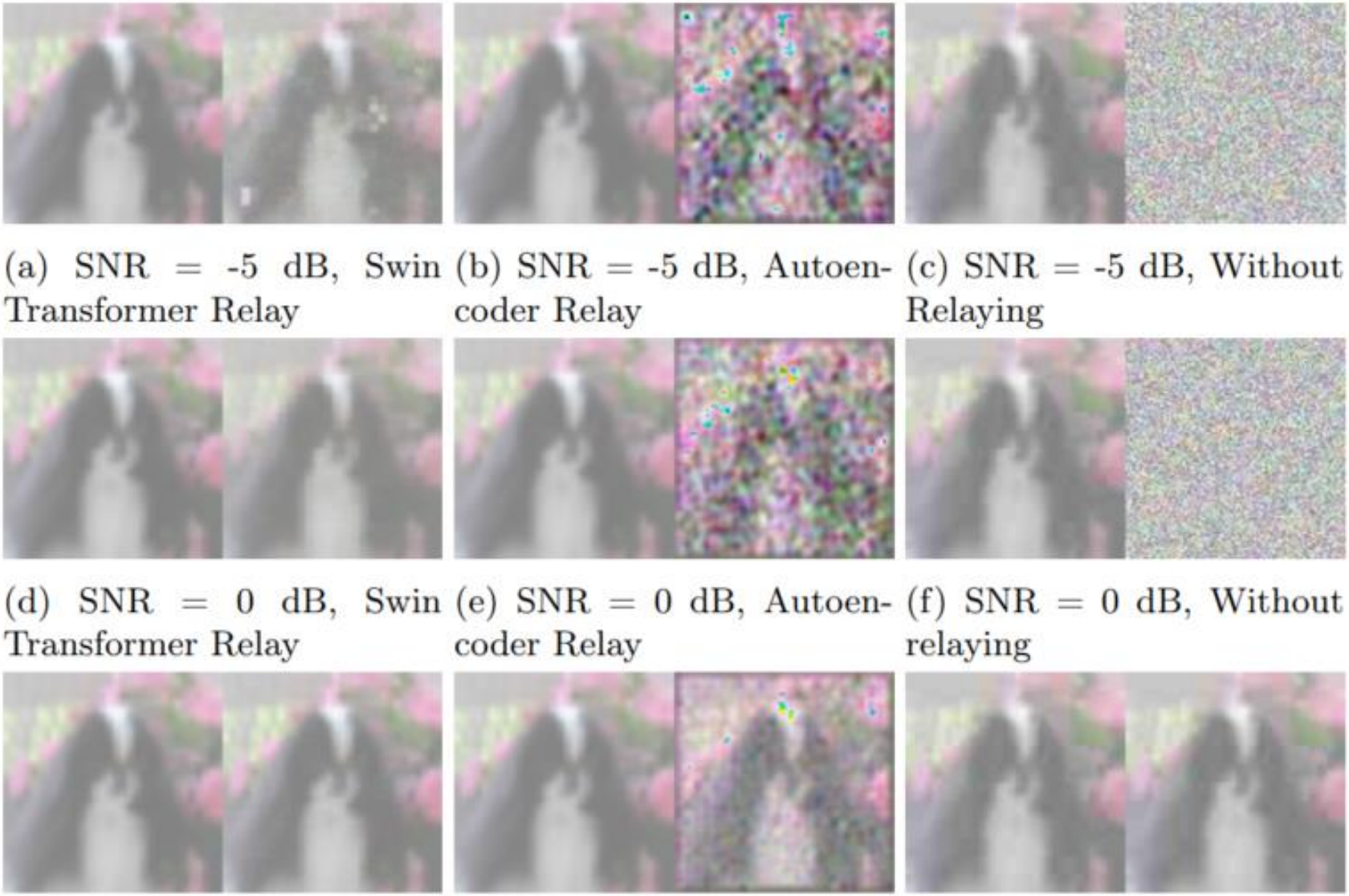


Secure Cooperative Semantic Communication with Swin Transformer and Jamming – Performance evaluation

CIFAR-10 dataset







(a) SNR = -5 dB, Swin Transformer Relay

(b) SNR = -5 dB, Autoencoder Relay

(c) SNR = -5 dB, Without Relaying

(d) SNR = 0 dB, Swin Transformer Relay

(e) SNR = 0 dB, Autoencoder Relay

(f) SNR = 0 dB, Without relaying

(g) SNR = 5 dB, Swin Transformer Relay

(h) SNR = 5 dB, Autoencoder Relay

(i) SNR = 5 dB, Without relaying

Fig. 7: Output images from cooperative systems with different types of relay.

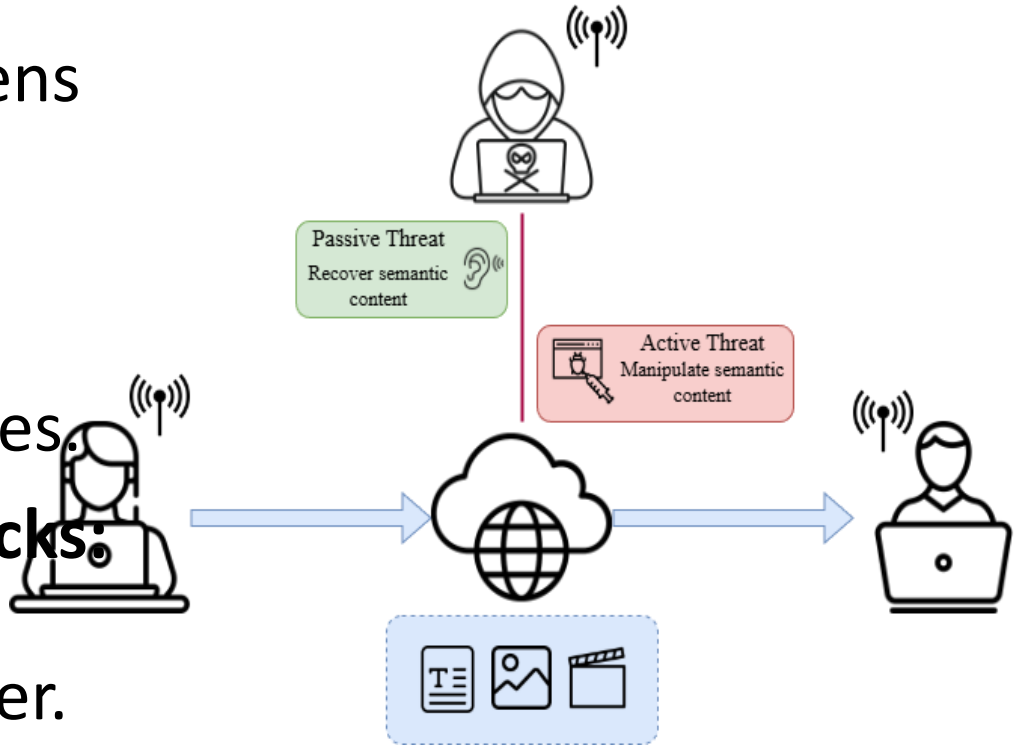


(a) SNR -5, Eavesdropper (b) SNR 0, Eavesdropper (c) SNR 5, Eavesdropper

Fig. 8: Reconstructed images at the Eve.

Vulnerabilities in SemCcom Systems

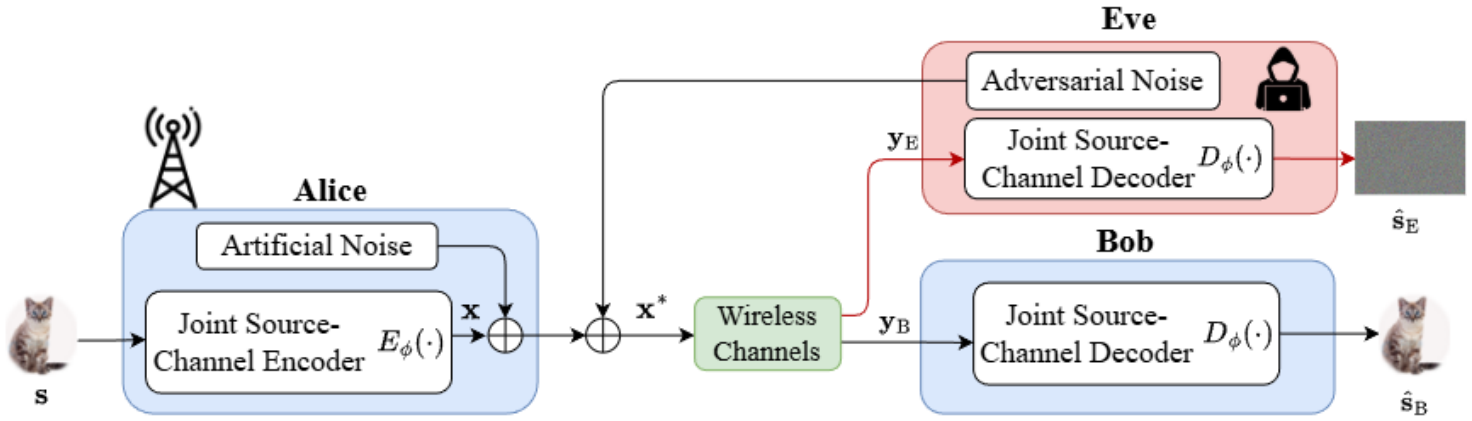
- **Shared Knowledge Risk:** Compromised shared knowledge base threatens communication.
- **Passive Eavesdropping:** Attackers intercept and decode semantic features.
- **Active Adversarial Attacks:** Crafted perturbations degrade/fool the receiver.



Secure Semantic Communications with Adversarial Training and Active Eavesdropper

Alice (Transmitter): Encodes the image via JSCC, incorporates adversarial training for robustness, and injects AN before transmission.

Eve (Active Adversary): Intercepts the transmitted signal and injects adversarial perturbations to disrupt communication.



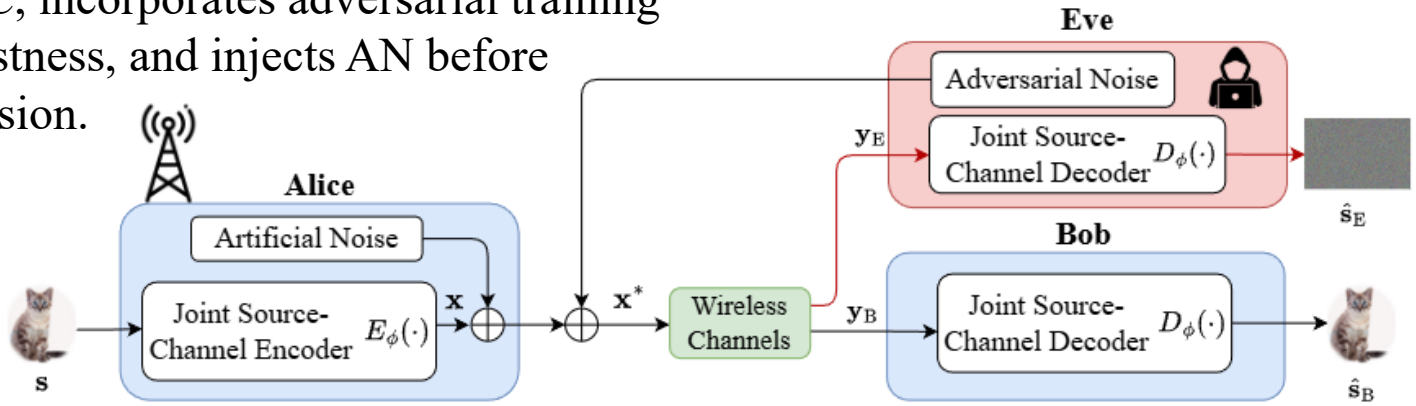
Bob (Legitimate Receiver): Removes the AN using the shared seed and accurately reconstructs the transmitted image.

Thai, Quang-Tinh, Toan-Van Nguyen, Trung-Nghia Nguyen, Tung Kieu, and Phuong Luu Vo. "Secure Semantic Communications with Adversarial Training and Active Eavesdropper." In *ACIIDS*, pp. 460-474, 2026.

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Quang-Tinh Thai, Toan-Van Nguyen, Trung-Nghia Nguyen, Tung Kieu, and Phuong Luu Vo. "Secure Semantic Communications with Adversarial Training and Active Eavesdropper." In *Asian Conference on Intelligent Information and Database Systems*, pp. 460-474. Singapore: Springer Nature Singapore, 2026.

Adversarial Training and Artificial Noise Strategy

Adversarial Training

Target Threat: Active adversarial perturbations injected into the wireless channel.

Mechanism:

- FGSM generates perturbations on transmitted features.
- Train on both **clean** and **attacked** samples.
- Optimize using min-max objective to improve robustness.

$$\min_{\theta,\phi} \mathbb{E}_{\mathbf{s}} \left[\max_{\|\delta\|_\infty \leq \epsilon} \mathcal{L}_{rec} \left(\mathbf{s}, \hat{\mathbf{s}}_B(\mathbf{x} + \delta) \right) \right]$$

Artificial Noise

Target Threat: Passive eavesdropping (unauthorized interceptions).

Mechanism:

- Inject pseudorandom AN into transmitted features.
- Bob: Removes AN using a shared random seed.
- Eve: Cannot remove AN, resulting in corrupted received features.

Simulation Settings

- Datasets:
  - Training: DIV2k (High-resolution images)
  - Testing: Kodak (unseen images)
- Training configuration:
  - Training Attacks: $\varepsilon = 0.01$
  - Testing Attacks: $\varepsilon = 0.1$
  - Attack Method: FGSM (White-box)
- Evaluation Metrics:
  - SNR Range: -15dB to 15dB.
  - Metrics: PSNR, SSIM, LPIPS, Clip_score, Euclidean Distance.
  - Key Comparison: Proposed SwinJSCC vs. Baseline vs. DeepJSCC

Image Quality Assessment Metrics

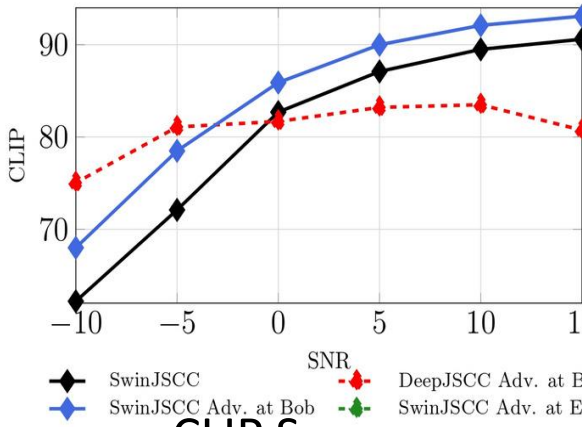
Signal Integrity

- **PSNR & MSE:** Measure pixel-level reconstruction accuracy.
- **SSIM**: Measures structural similarity between original and reconstructed images.
- **Euclidean Distance (ED):** Measures similarity between feature representations.

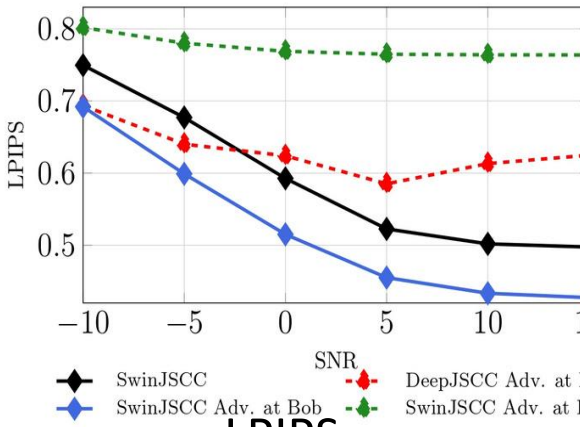
Semantic Fidelity

- **LPIPS (Learned Perceptual Image Patch Similarity):** Evaluates perceptual image quality based on human visual perception.
- **CLIP Score:** Measures semantic/content preservation of reconstructed images.

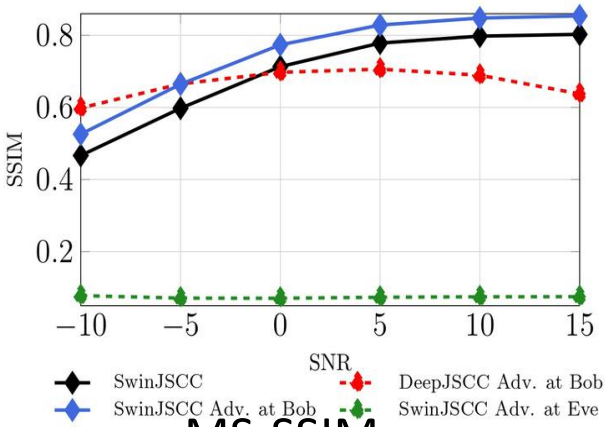
Results



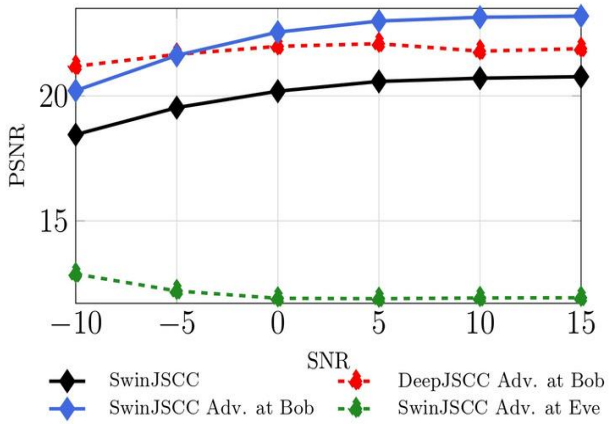
CLIP Score



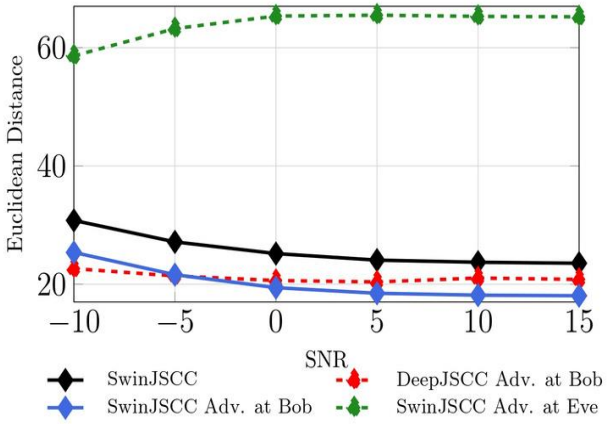
LPIPS



MS-SSIM

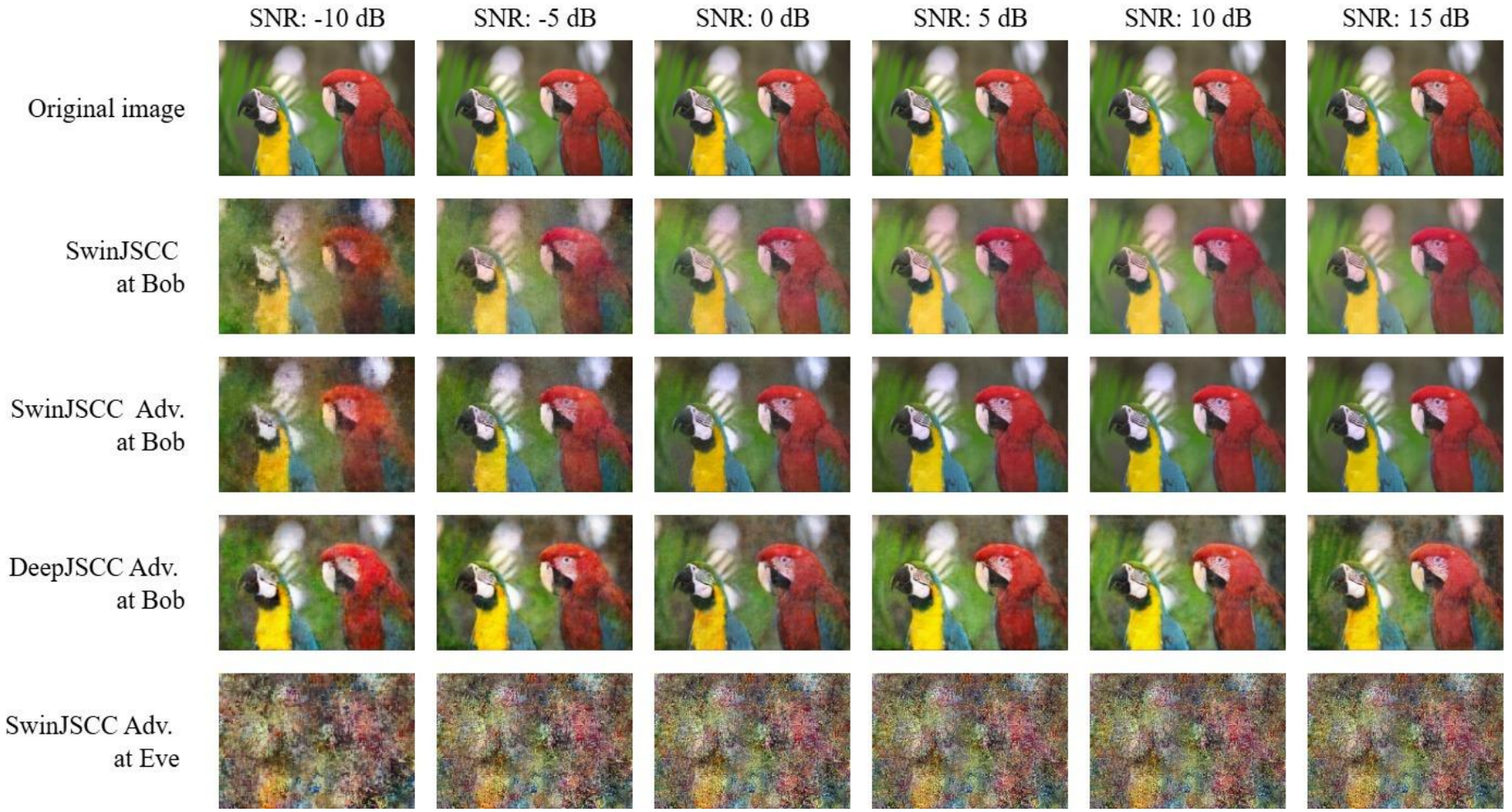


PSNR



ED

Results


SNR: -10 dB
SNR: -5 dB
SNR: 0 dB
SNR: 5 dB
SNR: 10 dB
SNR: 15 dB
Original image
SwinJSCC at Bob
SwinJSCC Adv. at Bob
DeepJSCC Adv. at Bob
SwinJSCC Adv. at Eve

Wasserstein Distributional Robust Optimization

Motivation

Standard ML (ERM)

The Core Assumption:

Standard models assume test data comes from the exact same probability distribution as training data (P_train = P_test).

The ERM Formula:

$$\min_{\theta} \left(\frac{1}{N}\right) \sum_{i}^{n} \ell(\theta, x_i)$$

The Flaw:

If real-world test conditions change even slightly, accuracy plummets drastically.

Real-World Shift

Out-of-Distribution Data:

Wireless channel fading, unexpected sensor noise, dynamic user movement, or corrupted signals.

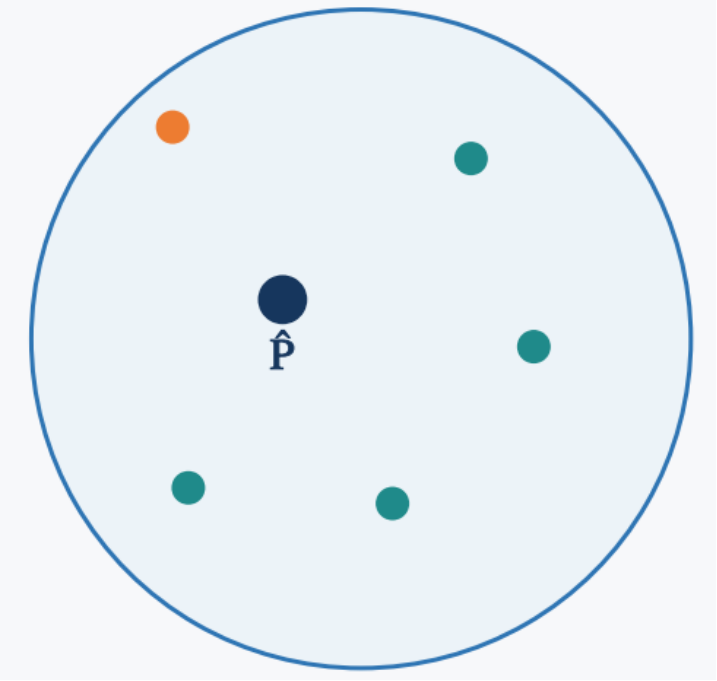
Brittle Model Behavior:

Models overfitted to training noise interpret unseen noise as meaningful patterns.

Catastrophic Failures:

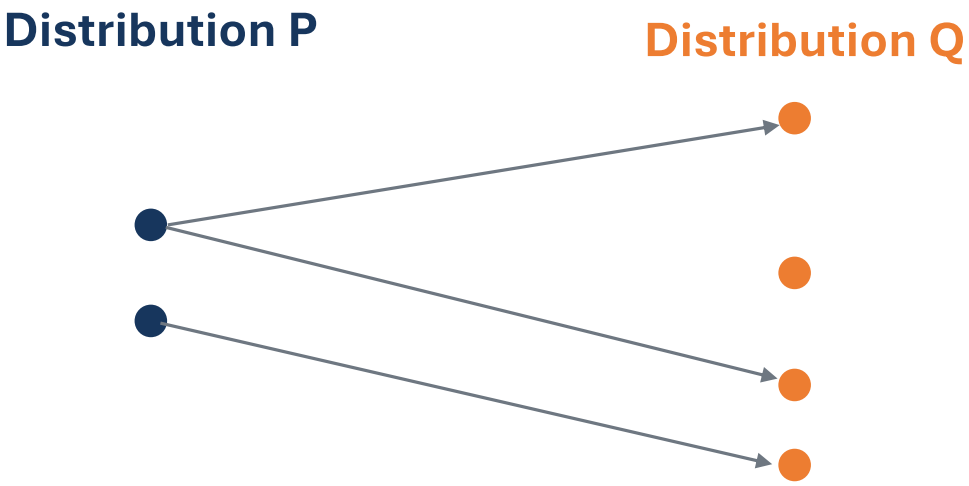
Standard deep networks give wrong predictions under shift.

Distributionally robust optimization (DRO)



- Learner chooses model parameters θ.
- Adversary chooses the worst plausible distribution Q

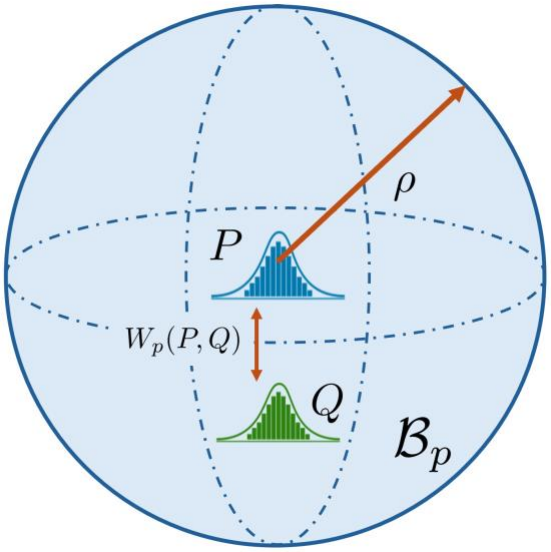
Wasserstein distance: “minimum work to move probability mass”



$$W_c(P, Q) = \inf_{\pi \in \Pi(P,Q)} E_\pi[c(z, z')]$$

- π is a transport plan coupling nominal samples z and perturbed samples z'. It tells us how mass from P is transported to Q.
- $d(z, z')$ is the distance between two points, e.g., $\|z - z'\|_2^2$
- The infimum chooses the cheapest transportation plan.

Wasserstein ball = all distributions within a transportation budget



$$\mathcal{B}_p(\mathrm{P},\rho) = \{Q : W_p(\mathrm{P},Q) \leq \rho\}$$

- $\rho = 0$: only the empirical distribution $\rightarrow$ ERM.
- small ρ: mild distribution shift.
- large ρ: stronger robustness, but potentially more conservative.

WDRO primal problem

$$\min_{\theta} \sup_{Q:W_c(\mathrm{P},Q)\leq\rho} \mathbb{E}_{z'\sim Q}\left[\ell(\theta; z')\right]$$

- Find model parameters that minimize the loss under the worst distribution that is not farther than ρ from the nominal distribution P.
- Inner problem is $\sup_{Q} \mathbb{E}_{Q}\left[\ell(\theta; z')\right]$, where Q is the entire probability distribution, not a finite dimensional vector ➔ **very difficult problem**.

Dual derivation of inner problem

- Worst Q:

$$\sup_Q \mathbb{E}_Q[\ell(\theta; z')]$$

the constraint is

$$W_c(P, Q) = \inf_{\pi \in \Pi(P,Q)} \mathbb{E}_{(z,z') \sim \pi}[c(z, z')] \leq \rho$$

Dual derivation

- Lagrange multiplier: $\lambda \geq 0$.
- Lagrangian function

$$L(Q, \lambda) = \mathbb{E}_Q[\ell(\theta; z')] - \lambda(W_c(P, Q) - \rho)$$
$$= \lambda\rho + \mathbb{E}_{\mathrm{Q}}[\ell(\theta; z')] - \lambda W_c(P, Q).$$

- Dual problem:

$$\min_{\lambda \geq 0} \left\{ \lambda\rho + \sup_{\mathrm{Q}} \left(\mathbb{E}_{\mathrm{Q}}[\ell(\theta; z')] - \lambda W_c(P, Q) \right) \right\},$$

Replace Wasserstein distance by its transport form

- Since

$$W_c(P,Q) = \inf_{\pi \in \Pi(P,Q)} \mathbb{E}_{(z,z')\sim\pi}\left[c(z,z')\right]$$

- Hence

$$\sup_{Q}\left(\mathbb{E}_{Q}[\ell(\theta;z')] - \lambda W_c(P,Q)\right) = \sup_{\pi}\mathbb{E}_{\pi}\left[\ell(\theta;z') - \lambda c(z,z')\right]$$

$$= \frac{1}{N}\sum_{i=1}^{N}\sup_{z'}[\ell(\theta;z') - \lambda c(z_i,z')]$$

- Distributional robust training:

$$\min_{\theta,\lambda\geq 0}\left\{\lambda\rho + \frac{1}{N}\sum_{i=1}^{N}\sup_{z'}[\ell(\theta;z') - \lambda c(z_i,z')]\right\}$$

Interpretation of the dual

- For each sample z_i, the adversary solves
$$z_i'^* = \underset{z'}{\operatorname{argmax}}[\ell(\theta; z') - \lambda c(z_i, z')]$$
- If λ is large, moving far away from z_i is expensive.
- Ifλ is small, stronger perturbations are permit
- We can apply the methods like gradient ascent in K steps of use first-order Taylor approximation to find $z_i'^*$.

Robust training algorithm

- Init θ, λ;

1. Sample a minibatch B;
2. Generate worst case samples by solving

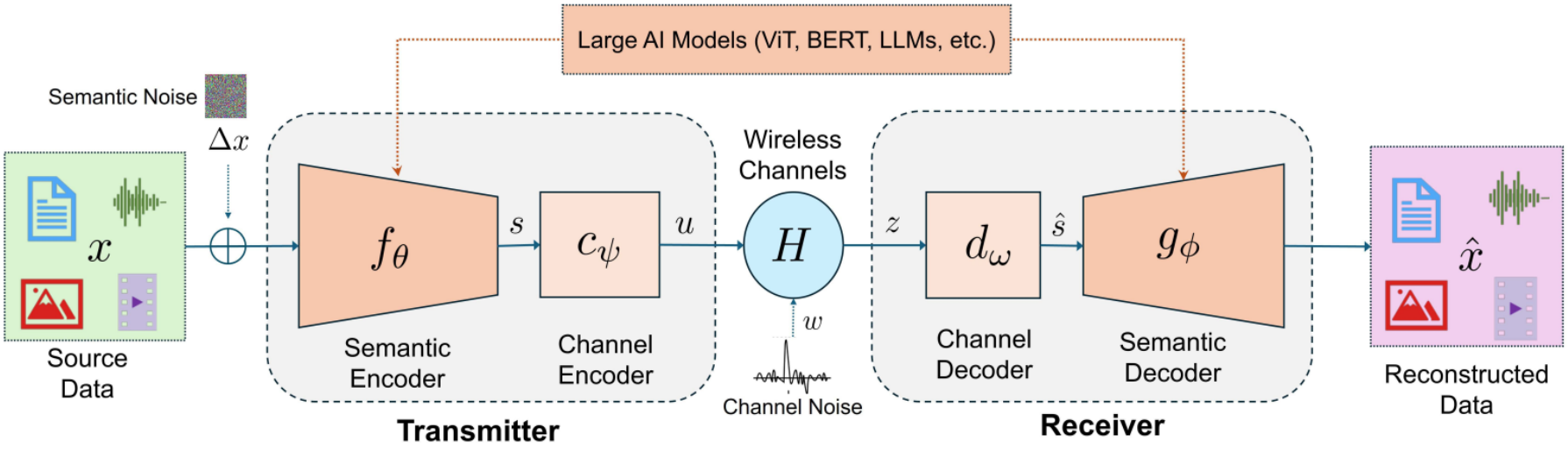
$$z_i'^{*} = \underset{z'}{\operatorname{argmax}}[\ell(\theta; z') - \lambda c(z_i, z')]$$

4. Calculate robust loss:

$$\lambda\rho + \frac{1}{\mathrm{B}}\sum_{i=1}^{B}[\ell(\theta; z_i'^{*}) - \lambda c(z_i, z_i'^{*})]$$

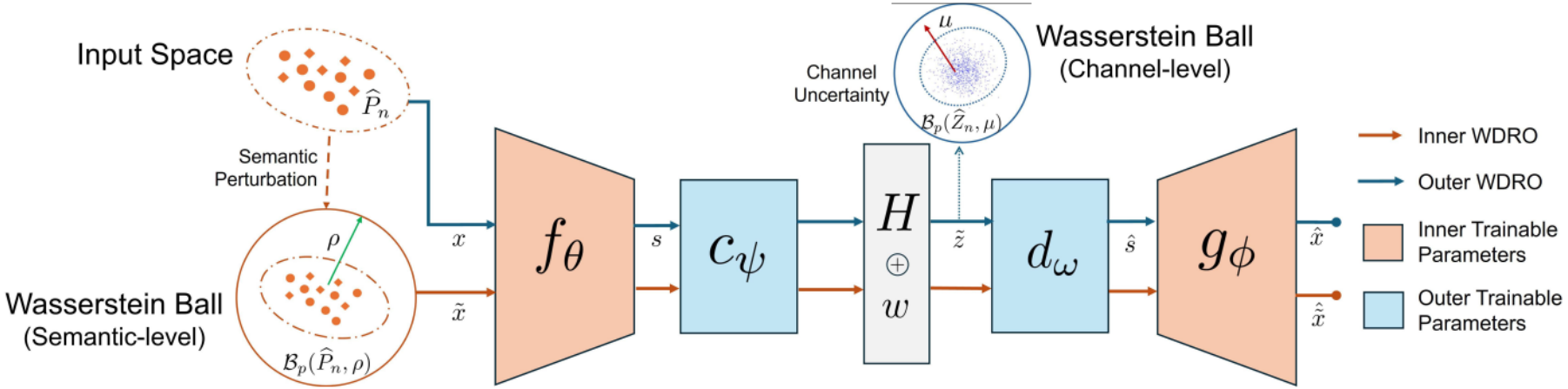
5. Update θ using gradient-based algorithm;
6. Update dual variable λ using projected gradient-based algorithm

Wireless semantic communication



- Source data is encoded into compact, task-relevant representations and transmitted over wireless channels;
- the receiver reconstructs the intended meaning, even under semantic and channel noise.

Overview of the WaSeCom framework



- The proposed bilevel WDRO model jointly optimizes semantic and channel encoder-decoder pairs for robustness.
- The inner level (orange) handles semantic input shifts, while the outer level (blue) addresses channel noise.

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Dual Form for Bi-level Problem

$$\text{INNER-DUAL:} \quad \min_{\theta,\phi} D(\theta,\phi|\psi,\omega), \text{ where } D(\theta,\phi|\psi,\omega) :=$$

$$\min_{\lambda\geq 0} \lambda\rho + \mathbb{E}_{x\sim\widehat{P}_n}\left[\sup_{\substack{\tilde{x}:z=h\,c_\psi(f_\theta(\tilde{x}))+w \\ \hat{\tilde{x}}=g_\phi(d_\omega(z))}} \{\ell_s(\tilde{x},\hat{\tilde{x}}) - \lambda c(x,\tilde{x})\}\Big|\psi,\omega\right] \tag{6}$$

$$\text{OUTER-DUAL:} \quad \min_{\psi,\omega} D(\psi,\omega|\theta^*), \text{ where } D(\psi,\omega|\theta^*) :=$$

$$\min_{\gamma\geq 0} \gamma\mu + \mathbb{E}_{z\sim\widehat{Z}_n}\left[\sup_{\substack{\tilde{z}:\hat{s}=d_\omega(\tilde{z}), \\ s=f_{\theta^*}(x),\, x\sim\widehat{P}_n}} \{\ell_c(s,\hat{s}) - \gamma\, c(z,\tilde{z})\}\,\Big|\theta^*\right]. \tag{7}$$

Require: Training data $\{x_i\}_{i=1}^n \sim \widehat{P}_n$, channel model h, noise model n; Wasserstein radii ρ, μ; no. of training steps T

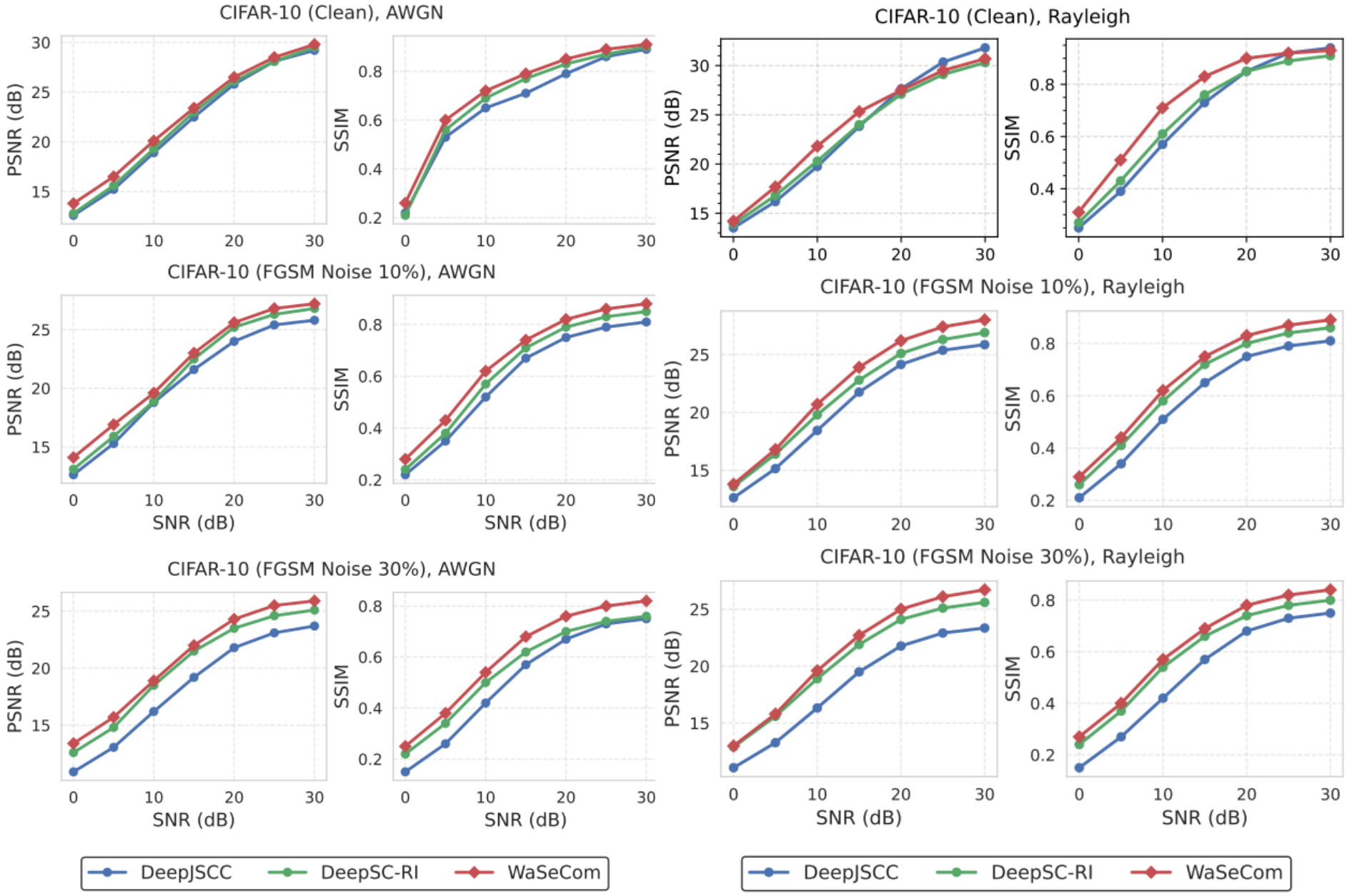
Initialize model parameters $\theta^{(0)}$, $\phi^{(0)}$, $\psi^{(0)}$, $\omega^{(0)}$ and dual variables $\lambda \geq 0$, $\gamma \geq 0$, $\epsilon \geq 0$
for $t = 1$ to T **do**
  Sample minibatch $\{x_i\}$ from $\widehat{P}_n$
  for OUTER LOOP **do**
    Compute semantic representation $s_i \leftarrow f_{\theta^{(t-1)}}(x_i)$
    Transmit through channel: $z_i \leftarrow h(c_{\psi^{(t-1)}}(s_i)) + w$
    Generate perturbed signals $\tilde{z}_i \in \mathcal{B}_p(z_i, \mu)$
    Decode received signal: $\hat{s}_i \leftarrow d_{\omega^{(t-1)}}(\tilde{z}_i)$
    Update $\psi^{(t)}$ and $\omega^{(t)}$ by solving problem (7) using gradient-based methods.
    for INNER LOOP **do**
      Generate semantic perturbation $\tilde{x}_i \in \mathcal{B}_p(x_i, \rho)$
      Compute encoded semantic $\tilde{s}_i \leftarrow f_{\theta^{(t-1)}}(\tilde{x}_i)$
      Compute channel output $z_i' \leftarrow h(c_{\psi^{(t)}}(\tilde{s}_i)) + w$
      Decode and reconstruct: $\hat{\tilde{x}}_i \leftarrow g_{\phi^{(t-1)}}(d_{\omega^{(t)}}(z_i'))$
      Update $\theta^{(t)}$ and $\phi^{(t)}$ by solving problem (6) using gradient-based methods.
    end for
  end for
end for

Experimens

Models:

- Semantic encoders:
  - Denoising Autoencoder Vision Transformer (DAE-ViT) for image
  - BERT-Base for text
- MLP-based channel encoder

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CIFAR-10 (Clean), AWGN
CIFAR-10 (Clean), Rayleigh
CIFAR-10 (FGSM Noise 10%), AWGN
CIFAR-10 (FGSM Noise 10%), Rayleigh
CIFAR-10 (FGSM Noise 30%), AWGN
CIFAR-10 (FGSM Noise 30%), Rayleigh
PSNR (dB)
SSIM
SNR (dB)
DeepJSCC
DeepSC-RI
WaSeCom

SNR (dB)	Noise-Free			FGSM Noise 10%			FGSM Noise 30%		
	DeepJSCC	DeepSC-RI	WaSeCom	DeepJSCC	DeepSC-RI	WaSeCom	DeepJSCC	DeepSC-RI	WaSeCom
0	12.61	12.80	13.80	12.63	13.10	14.10	10.91	12.60	13.40
5	15.22	15.60	16.50	15.29	15.90	16.90	13.06	14.80	15.70
10	18.91	19.30	20.10	18.79	18.90	19.60	16.20	18.50	18.90
15	22.50	23.00	23.40	21.60	22.50	23.00	19.20	21.50	22.00
20	25.80	26.10	26.50	24.00	25.20	25.60	21.80	23.50	24.30
25	28.10	28.10	28.50	25.40	26.30	26.80	23.10	24.60	25.50
30	29.20	29.50	29.80	25.80	26.80	27.20	23.70	25.10	25.90

TABLE I: PSNR (dB) comparison under Clean and FGSM attacks over the AWGN channel on CIFAR-10.

SNR (dB)	Noise-Free			FGSM Noise 10%			FGSM Noise 30%		
	DeepJSCC	DeepSC-RI	WaSeCom	DeepJSCC	DeepSC-RI	WaSeCom	DeepJSCC	DeepSC-RI	WaSeCom
0	13.52	13.90	14.20	12.63	13.60	13.80	11.08	12.90	13.00
5	16.19	16.80	17.68	15.15	16.40	16.80	13.28	15.60	15.80
10	19.76	20.30	21.81	18.45	19.80	20.70	16.33	18.90	19.60
15	23.83	24.00	25.32	21.76	22.80	23.90	19.50	21.90	22.70
20	27.64	27.10	27.65	24.15	25.10	26.20	21.77	24.10	25.00
25	30.36	29.10	29.50	25.38	26.30	27.40	22.91	25.10	26.10
30	31.80	30.30	30.70	25.86	26.90	28.00	23.35	25.60	26.70

TABLE II: PSNR (dB) comparison under Clean and FGSM attacks over the Rayleigh channel on CIFAR-10.

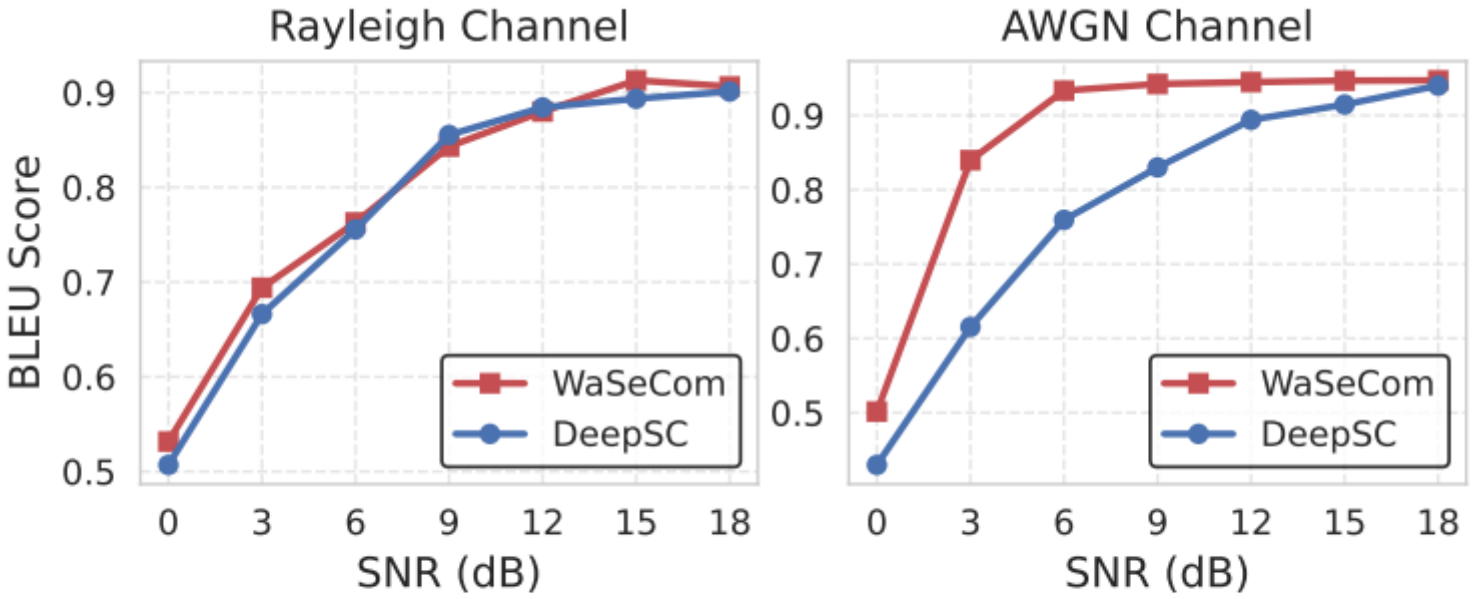


Fig. 6: Performance of text transmission without semantic noise under different channels

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Open question

- Using Sinkhorn distance instead of Wasserstein distance
- Wang, Jie, Rui Gao, and Yao Xie. "Sinkhorn distributionally robust optimization." *Operations Research* 74, no. 3 (2026): 1581-1603.

References

- https://adversarial-ml-tutorial.org/